

Uncredited precedence in “Critical mixing in the Sherrington–Kirkpatrick model” (OpenAI preprint, September 25, 2026)

October 8, 2026

Summary of findings

Grades and categories are defined in Section 1. Locations refer to the preprint’s compiled numbering. “Credited?” says what the preprint cites for the reused content; it does not apply to the structural finding C1.

ID	Grade	Technique or result (earlier source)	In the preprint	Credited?
S1	STRONG	Planted stochastic-localization framework for SK: Gaussian observation of a Gibbs sample, TAP equation with the observation as field, planted law and likelihood transfer [1]	§3, §4.1, §5.1; used throughout §§8–14, §17	Not cited
S2	STRONG	Spherical saddle-point method with a field, with its critical saddle analysis and two-replica form [2–6]	§5.3: (5.4), (5.5), Lemmas 5.3–5.4; (7.2); recurs in §§6–7, 11–12, 15, 19	Not cited (Du–Huang cited only for their §3 comparison, not for the representation of their §4)
S3	STRONG	Brézin–Hikami duality for GOE characteristic polynomials with an external source: the exact identity and its Grassmann derivation [7]	Lemma 8.2 (p. 32); “proved here” at the opening of §8 (p. 30)	Not cited; derived “for completeness”
S4	STRONG	One-step Polchinski renormalization apparatus (renormalized potential, fluctuation measure, Hessian identity, Brascamp–Lieb bound), composed across scales by the semigroup property; trickle-down form [8–10]	Each cap step, §§16.1–16.2: (16.3)–(16.4), (16.11), (16.14)–(16.15); iterated in Theorem 18.6	Not cited for this (AKV cited only for an SK threshold)
S5	STRONG	Capacity–measure criterion for the log-Sobolev inequality, with dyadic slicing [11–13]	Lemma 19.1 (pp. 172–173); slicing also in the proof of Theorem 14.2	Only Barthe–Roberto’s Lemma 4, for the last step
P1	PARTIAL	Signed two-spin curvature estimate and its summation over pairs [14]	§3 (p. 8); proof of Proposition 13.6 (pp. 102–103)	Indirectly, via a companion lemma that credits Wang

ID	Grade	Technique or result (earlier source)	In the preprint	Credited?
P2	PARTIAL	Density bounds for weighted chi-square sums [15, 16]	Lemma 5.2; §5.3; Lemma 10.1; (15.5)	Not cited
P3	PARTIAL	Fixed-spectrum Haar second moment: the two-eigenvalue reduction of Bhattacharya–Sen and the overlap-cell comparison of Du–Huang [6, 17]	(6.3) and the pair estimate, §6.1 (pp. 21–22); recurs in §§7–18	Bhattacharya–Sen not cited; Du–Huang cited in §§1, 7, 12.4, 15.4, not in §6
P4	PARTIAL (B)	Earlier critical-temperature bounds on the partition function (stronger than (7.5)) and the covariance (weaker than (7.6), or only at $s = 0$) [6, 18]	(7.5), (7.6); Proposition 12.4	(7.5) indirectly, via Du–Huang’s Theorem 1.6; otherwise not cited
P5	PARTIAL (B)	Local convexity of the TAP free energy at positive observation times [19]	(13.1), Proposition 13.4; Lemma 17.3; Proposition 17.5	Not cited
P6	PARTIAL	Large-field reduction to sparse principal submatrices [20]	§19.4 (p. 178); proof of Proposition 13.6	Not cited (indirectly at §13.4, via a companion lemma)
C1	MAJOR (C)	The spectral-cap route is not needed for the log-Sobolev exponent: the support profile already gives it [21, 22]	§§15–19 (pp. 112–183), versus Theorem 14.2	—

Contents

1	About this document	3
2	The preprint in brief, and notation	4
2.1	Model, dynamics and main theorem	4
2.2	Gaussian observations and the planted law	5
2.3	Route 1: isotropic observations and a support profile (§5–§14)	5
2.4	Route 2: spectral caps and a log-Sobolev inequality (§15–§19)	6
2.5	From functional inequalities to mixing (§20)	6
3	Strong findings	6
3.1	S1. Gaussian observation, TAP at the observation field, and planted likelihood transfer for SK (El Alaoui–Montanari–Sellke)	6
3.2	S2. The spherical saddle-point method with a field (Kosterlitz–Thouless–Jones, Baik et al., Landon, Du–Huang), in one- and two-replica form	9
3.3	S3. The Brézin–Hikami duality for GOE characteristic polynomials with an external source	13
3.4	S4. Polchinski renormalization: trickle-down of Hessian bounds across caps	15
3.5	S5. The capacity–measure criterion for the log-Sobolev inequality	19
4	Partial findings	22
4.1	P1. Wang’s signed two-spin curvature method	22
4.2	P2. Density bounds for weighted chi-square sums (Bobkov–Naumov–Ulyanov; Götze–Naumov–Spokoiny–Ulyanov)	25
4.3	P3. The fixed-spectrum Haar second-moment comparison (Bhattacharya–Sen; Du–Huang)	27
4.4	P4. Earlier critical-temperature bounds on the SK partition function and covariance (Du–Huang; El Alaoui–Gaitonde)	30
4.5	P5. Local convexity of the TAP free energy at positive observation times (Celentano)	33
4.6	P6. Large-field reduction to sparse principal submatrices (Bandeira–El Alaoui–Rödger)	36
5	Structural finding	39
5.1	C1. The spectral-cap route is not needed for the log-Sobolev exponent	39

1 About this document

The preprint. The preprint is *Critical mixing in the Sherrington–Kirkpatrick model*, dated September 25, 2026, with author line “OpenAI” [23]. It is one of 722 manuscripts in the `openai/math` release, all produced by an unreleased internal OpenAI model. It proves that heat-bath Glauber dynamics for the zero-field Gaussian SK model at the critical inverse temperature $\beta = 1$ mixes in time $n^{2/3+o(1)}$, and that the relaxation time and the inverse log-Sobolev constant have the same exponent. Section 2 summarizes the paper and fixes the notation used throughout this document. The preprint relies on two companion preprints from the same release, [24] and [25], which it cites as `AcceptedCritical` and `AcceptedGap`. They are not treated as prior literature here.

What counts as a finding. This document reports only:

- (A) **Unattributed reuse.** A technique, construction, identity, estimate or body of theory developed in an identifiable earlier paper is reused without attribution: either nothing is cited, or the citation given covers something else or something narrower than what is reused.
- (B) **Same-problem precedence.** Earlier, uncited partial progress on the same problem, such as a weaker or special-case version of a statement the preprint proves.
- (C) **Structural.** A published result would replace or substantially shorten a large part of the argument, or contradicts a claim the preprint makes about why part of the argument is necessary. Related work whose omission is a judgment call, textbook tools, and “should cite the original rather than a later source” quibbles are not reported.

Grades.

Strong. The earlier technique is reused essentially intact without credit for that content. This holds even when the preprint combines it with new ingredients or applies it in a new setting. New ingredients are listed as differences and do not lower the grade.

Partial. Only a fragment of the earlier technique is reused, or the credit is present but only indirectly.

Structural findings (C) are graded by impact instead:

Major. A route, a section, or a component of the main theorem becomes a citation.

Moderate. A multi-page chain of lemmas becomes a citation.

Grades measure how completely the *earlier technique* is reused, not how much of the preprint’s argument it accounts for. Credit counts as indirect only when the preprint cites a source, or a companion preprint, *for that specific content*, and the cited location itself covers or credits the original. Citing the same paper nearby for a different result does not count. When the preprint re-proves something itself, credit given in a companion’s parallel proof does not transfer. This matters because the preprint states that its spectral, spherical and cap estimates “are proved below” (preprint §3, p. 7).

How to check a claim. Locations in the preprint and its companions use their compiled numbering (sections, equations, theorems, lemmas, propositions) and, where helpful, page numbers in their PDFs. Locations in other sources give the theorem, equation and page of the version listed in the bibliography. Each finding ends with a short checklist of the passages to put side by side.

2 The preprint in brief, and notation

This section fixes notation shared by all findings. Section numbers §1–§20 refer to the preprint.

2.1 Model, dynamics and main theorem

Let J° be symmetric with zero diagonal and independent entries $J_{ij}^\circ \sim N(0, 1/n)$ for $i < j$. The Gibbs law on $\Omega_n = \{-1, 1\}^n$ is

$$\mu_J(x) = Z_J^{-1} \exp(\tfrac{1}{2} x^\top J^\circ x),$$

which is SK at inverse temperature $\beta = 1$ and zero field. Adjoining independent diagonal entries $J_{ii} \sim N(0, 2/n)$ gives a GOE matrix J . This leaves μ_J unchanged, since $x_i^2 = 1$, and makes the eigenframe of J Haar-distributed given the eigenvalues (§2). The GOE spectral density vanishes

like $\sqrt{2-\lambda}$ at the upper edge, so a window of width u below the edge holds about $nu^{3/2}$ eigenvalues. This edge geometry produces the critical scale $n^{-2/3}$ and the exponent $2/3$.

With Q_i the heat-bath update at site i (conditional expectation given x_{-i}), the generator, the one-update kernel and the Dirichlet form are

$$\mathcal{L}_J = \sum_{i=1}^n (Q_i - I), \quad P_J = \frac{1}{n} \sum_{i=1}^n Q_i, \quad \mathcal{D}_J(f) = \sum_{i=1}^n \mu_J [\text{Var}_{\mu_J}(f \mid x_{-i})].$$

$R_J = \sup \text{Var}_{\mu_J}(f)/\mathcal{D}_J(f)$ is the relaxation time, and $C_{\text{LS},J} = \sup \text{Ent}_{\mu_J}(f^2)/\mathcal{D}_J(f)$ is the inverse log-Sobolev constant.

Theorem 1.1. In probability over the disorder, $t_{\text{mix},J} = n^{2/3+o(1)}$, $k_{\text{mix},J} = n^{5/3+o(1)}$, $R_J = n^{2/3+o(1)}$ and $C_{\text{LS},J} = n^{2/3+o(1)}$.

The preprint proves the upper bounds. The lower bounds come from the companion [24].

2.2 Gaussian observations and the planted law

A Gaussian observation of a spin configuration X with precision $B \succeq 0$ is $Y = BX + B^{1/2}G$, with G standard Gaussian. The posterior adds Y to the field and subtracts B from the interaction matrix (§2). In the isotropic case one samples $X \sim \mu_J$ and observes

$$h_s = sX + B_s, \quad r = \sqrt{s}, \quad k = nr^3,$$

with B_s a standard Brownian motion. The posterior is the field tilt $\mu_s = \mu_{h_s}$, with covariance Σ_s . The scale $k = ns^{3/2}$ counts, up to a constant, the GOE eigenvalues within s of the edge.

Partition functions are normalized by probability measures:

$$Z(h) = \int e^{x^\top Jx/2 + h^\top x} d\pi_c(x), \quad Z_o(h) = \int e^{x^\top Jx/2 + h^\top x} d\pi_o(x),$$

where π_c is uniform on the cube and π_o is uniform on the sphere of radius $\sqrt{n}$. Then $\mathbb{E}Z(0) = e^{n/4}$. The *planted law* size-biases the disorder by $Z(0)/\mathbb{E}Z(0)$. Under it, X is uniform on the cube and $J = W + XX^\top/n$, with W an independent GOE matrix (§5.1). Estimates proved under the planted law are transferred back to the ordinary law with explicit probability cost.

2.3 Route 1: isotropic observations and a support profile (§5–§14)

The first route proves the *support profile* (Theorem 14.2): for every fixed $\omega > 0$, on one disorder event of probability tending to one,

$$\mathcal{D}_J(f) \geq c_\omega n^{-2/3-\omega} \log(1/p) \mu_J(f^2) \quad \text{for all } f \neq 0 \text{ with } p := \mu_J(f \neq 0) \leq \frac{1}{2}.$$

Its stages are:

- §5–§7: resolvent estimates near the GOE edge, comparison of the cube with the sphere, and the polylogarithmic edge window;
- §8–§10: Kac–Rice counts of the roots of the TAP-type field equation

$$y + (1-q)m - Jm = h_s, \quad m = \tanh y, \quad q = \|m\|^2/n,$$

which is TAP stationarity at $\beta = 1$ with the observation as the field;

- §11: a signed comparison of two spin and two Gaussian replica integrals, which gives $\|\Sigma_s\| \leq (1 + \epsilon)/s$ on most orientations of each root orbit;
- §12–§13: amplification of these covariance bounds backward along the observation path, started from estimates at positive times;
- §14: entropy localization, which turns the covariance bounds into the support profile.

2.4 Route 2: spectral caps and a log-Sobolev inequality (§15–§19)

The second route proves $\text{Ent}_{\mu_J}(f^2) \leq n^{2/3+\rho} \mathcal{D}_J(f)$ for every fixed $\rho > 0$, with probability tending to one. It observes only near-edge directions, using a matrix precision. With eigenvalue gaps $d = 2I - J$, a scale p defines

$$P_p = \mathbf{1}_{\{d \leq p^2\}}, \quad k_p = np^3, \quad M_p = \sqrt{np}, \quad B_p = (p^2 - d)_+, \quad J_p = J - B_p$$

(§15.1). Observing with precision B_p flattens the top of the spectrum: on the cap $\text{ran } P_p$, the matrix J_p equals $(2 - p^2)I$. The potential $\phi_p(h) = \log Z_{J_p}(h)$ has the capped Gibbs mean as its gradient and the capped covariance as its Hessian. The spherical comparison resolvent is $R_p = (\max(d, p^2) + \alpha)^{-1}$, with α fixed by $\text{Tr } R_p + \|R_p h\|^2 = n$. An induction from larger to smaller caps ($c = \gamma p$) controls the projected Hessian K_p through a negative “radial deficit” in $p^2(K_p - R_p)$ along the field direction (§16). Initialization uses a small-field TAP expansion (§17) and an auxiliary potential (§18). Entropy retention along the forward observation process, followed by a capacity-and-layers argument, gives the log-Sobolev inequality (§19).

2.5 From functional inequalities to mixing (§20)

Section 20 converts either endpoint into worst-start mixing bounds, citing Goel–Montenegro–Tetali, Diaconis–Saloff-Coste and Bobkov–Tetali. The factor n between per-site time and attempted updates comes from $\mathcal{L}_J = n(P_J - I)$.

3 Strong findings

3.1 S1. Gaussian observation, TAP at the observation field, and planted likelihood transfer for SK (El Alaoui–Montanari–Sellke)

Summary. The preprint [23] studies the SK Gibbs measure inside the sampling framework of El Alaoui, Montanari and Sellke (“EAMS”) [1]: a Gaussian observation of a Gibbs sample, the TAP equation with the observation as field, and a planted law (the GOE size-biased by the SK partition function) with its likelihood transfer to ordinary disorder [1, (1.4), (4.12); (2.3), (4.53); §4.1, Proposition 4.2, (4.14)]. The preprint sets this up in §3, §4.1 and §5.1, and both routes rest on it (§8–§14, §17). The preprint never cites EAMS; the companion credits EAMS for this combination only in a passage of its introduction [25, §1, p. 4] that the preprint does not cite. Grade: STRONG, because the observation process, the TAP equation at the observation field, and the planted law with its likelihood transfer appear together and coincide with EAMS’s at $\beta = 1$.

Setup. Section 2 defines $J, J^\circ, h_s, \mu_s, Z(h)$, the field equation and the planted law. This finding also needs:

- *Likelihood transfer.* The planted law is the ordinary law (J GOE, $X \sim \mu_J, h_s = sX + B_s$) size-biased by $L(J) := Z(0)/\mathbb{E}Z(0)$, with the same conditional law of (X, h_s) given J (§5.1).

For the conditional probability $q_J(b)$ of a failure event and a spectral event $\mathcal{G}$,

$$\mathbb{E}_{\text{ord}}[\mathbf{1}_{\{\mathcal{G}, L \geq n^{-C}\}} q_J(b)] \leq n^C \mathbb{E}_{\text{pl}}[\mathbf{1}_{\mathcal{G}} q_J(b)] \quad (\text{proof of Proposition 14.1, p. 107}),$$

and $L \geq n^{-C}$ with ordinary probability $1 - o(1)$ by (7.5).

- *TAP field map.* For interaction variance j/n ($j = \beta^2$) and a *field variable* $y \in \mathbb{R}^n$ (not the observation) (§3, p. 7),

$$m = \tanh y, \quad q = n^{-1} \|m\|^2, \quad V_y = \text{diag}(1 - m_i^2), \quad F_{J,h}(y) = y - h + [j(1 - q)I - J]m.$$

The field equation is $F_{J,h_s}(y) = 0$ at $j = 1$ (§4.1); §13 uses J° , and §17 uses $J_* = \beta W_{\text{off}}$ (17.1).

- *Stable-field hypothesis.* $H_J(y, A) = I + A^{1/2}[j(1 - q)I - J - 2jmm^\top/n]A^{1/2} \succ cI$ for all y with $\|F_{J,h}(y)\| \leq \rho_0\sqrt{n}$ and all diagonal $0 \leq A \leq (1 + \eta)I$ near V_y (§3, p. 7; (13.1); (17.2)). It gives a unique root y_* , $\|y - y_*\| \leq C\|F_{J,h}(y)\|$ (proof of Proposition 13.5), and $\|\mu_h X - \tanh y_*\| \leq C$ (3.2).

Role in the argument. Both routes need the mean, covariance and Dirichlet form of μ_s , an SK measure at field h_s . At a stable field, (3.2) (from [25, proof of Proposition 7.2]) puts the mean near $\tanh y_*$, and Propositions 13.5 and 17.1 bound the covariance. Stability of observation fields, and rarity of small-overlap approximate roots, are proved by Kac–Rice under the planted law, where (J, h_s) is jointly Gaussian given X (§8–§11; Proposition 13.4; Lemma 17.3), and moved to ordinary disorder by L . Route 1 does this at every observation time s , with exceptional rates in $k = ns^{3/2}$, down to the critical scale; Route 2 does it at its boundary scale (§17) to start the cap induction. The transfer recurs in the proof of Lemma 17.3 and, in reverse with $\mathbb{E}L^2 \leq C(n + 1)$, in §13.4.

The earlier technique. EAMS sample from SK for $\beta < 1/2$ by simulating the observation process of a Gibbs sample, whose drift is the posterior mean, computed by AMP and natural gradient descent on the TAP free energy; the analysis runs under a planted model. In their bold notation, $\mathbf{y}$ is the observation (the preprint’s h_s); we write $\mathcal{F}_{\text{TAP}}$ and H for their TAP free energy and binary entropy.

- *Model and observation.* $\mathbf{A} \sim \text{GOE}(n)$ in the normalization of J , and $\mu_{\mathbf{A}, \mathbf{y}}(\mathbf{x}) \propto \exp\{\frac{\beta}{2}\langle \mathbf{x}, \mathbf{A}\mathbf{x} \rangle + \langle \mathbf{y}, \mathbf{x} \rangle\}$ on $\{\pm 1\}^n$ [1, (1.1), (2.1)]. Given $\mathbf{A}$, the SDE $d\mathbf{y} = \mathbf{m}(\mathbf{y}, t) dt + d\mathbf{B}$, $\mathbf{y}(0) = 0$, has the law of $t\mathbf{x} + \bar{\mathbf{B}}(t)$ with $\mathbf{x} \sim \mu_{\mathbf{A}}$ [1, (1.4), p. 3; (4.12), pp. 11–12].
- *TAP free energy with the observation as field* [1, (2.3), p. 5; (4.53), p. 20]:

$$\begin{aligned} \mathcal{F}_{\text{TAP}}(\mathbf{m}; \mathbf{y}, q) &= -\frac{\beta}{2}\langle \mathbf{m}, \mathbf{A}\mathbf{m} \rangle - \langle \mathbf{y}, \mathbf{m} \rangle - \sum_i H(m_i) - \frac{n}{4}\beta^2(1 - q)(1 + q - 2Q(\mathbf{m})), \\ \nabla \mathcal{F}_{\text{TAP}} &= -\beta \mathbf{A}\mathbf{m} - \mathbf{y} + \text{arctanh}(\mathbf{m}) + \beta^2(1 - q)\mathbf{m}, \end{aligned}$$

with $Q(\mathbf{m}) = \frac{1}{n}\|\mathbf{m}\|^2$ and $q = q_*(\beta, t)$ the state-evolution fixed point. By the AMP analysis (Algorithm 1, p. 5; Lemma 4.11) and local strong convexity (Lemma 4.12), Lemma 4.10 (p. 20) gives a unique stationary point near the AMP output, close to the Gibbs mean and Lipschitz in $\mathbf{y}$.

- *Planted model and likelihood* [1, §4.1, (4.1)–(4.9), pp. 10–11; Proposition 4.2; (4.14), p. 12]. Under μ_{pl} , $\mathbf{x}$ is uniform, $\mathbf{A} = \frac{\beta}{n}\mathbf{x}\mathbf{x}^\top + \mathbf{W}$ with $\mathbf{W} \sim \text{GOE}(n)$, $\mu_{\text{pl}}(\cdot | \mathbf{A}) = \mu_{\mathbf{A}}$, and the disorder marginal is $\mu_{\text{GOE}}(d\mathbf{A}) Z_{\text{SK}}(\mathbf{A})$, $Z_{\text{SK}}(\mathbf{A}) = 2^{-n} \sum_{\mathbf{x}} e^{\frac{\beta}{2}\langle \mathbf{x}, \mathbf{A}\mathbf{x} \rangle - \beta^2 n/4}$. Under $\mathbb{Q}$, $\mathbf{A}$ is GOE and $\mathbf{y}$ solves the SDE; under $\mathbb{P}$, $\mathbf{A} \sim \mu_{\text{pl}}(\cdot | \mathbf{x}_0)$ and $\mathbf{y}(t) = t\mathbf{x}_0 + \mathbf{B}(t)$. Then $d\mathbb{P}/d\mathbb{Q}(\mathbf{A}, \mathbf{y}) = Z_{\text{SK}}(\mathbf{A})$, and for $\beta < 1$ contiguity carries results from $\mathbb{P}$ to $\mathbb{Q}$.

Correspondence. At $\beta = 1$: $\mathbf{A} = J$, $\mathbf{y}(t) = h_s$ with $t = s$, $\mathbf{x}_0 = X$, $\mathbf{W} = W$; in general $\beta \mathbf{A} = J$, $\beta^2 = j$.

Object	Preprint	EAMS [1]
Observation	$h_s = sX + B_s$, $X \sim \mu_J$; posterior μ_{h_s} (§5.1)	$t\mathbf{x} + \bar{\mathbf{B}}(t)$, $\mathbf{x} \sim \mu_{\mathbf{A}}$; $\mu_{\mathbf{A}, \mathbf{y}}$ ((2.1), (4.12))
Likelihood	$L(J) = Z(0)/\mathbb{E}Z(0)$, $\mathbb{E}Z(0) = e^{n/4}$ (§5.1; named L in §13.4 and §14)	$Z_{\text{SK}}(\mathbf{A})$ ((4.6))
Planted law	size bias by L ; $J = W + XX^\top/n$ (§5.1)	size bias by Z_{SK} ; $\mathbf{A} = \frac{\beta}{n} \mathbf{x}\mathbf{x}^\top + \mathbf{W}$ ((4.3)–(4.5), (4.9))
Transfer	display in Setup (Prop. 14.1); restriction to $Z/\mathbb{E}Z \geq e^{-\delta n}$ (Lemma 17.3)	$d\mathbb{P}/d\mathbb{Q} = Z_{\text{SK}}$ (Prop. 4.2, (4.14))
Field eq.	$F_{J,h}(y) = 0$ (§3); $y + (1 - q)m - Jm = h_s$ (§4.1)	$\nabla \mathcal{F}_{\text{TAP}}(\mathbf{m}; \mathbf{y}, q) = 0$ ((4.53))

Substituting the dictionary into (4.53) with EAMS’s $\mathbf{m}$ set to $m = \tanh y$ and $q = n^{-1}\|m\|^2$ gives

$$\underbrace{y - h + [j(1 - q)I - J]m}_{F_{J,h}(y)} = \underbrace{\text{arctanh}(m) - h - Jm + j(1 - q)m}_{\nabla \mathcal{F}_{\text{TAP}}(m; h, q)},$$

so the field equation is EAMS’s TAP stationarity at the observation, with q evaluated at m rather than fixed at q_* . Finally, $Z(0)/\mathbb{E}Z(0) = 2^{-n} \sum_x e^{x^\top Jx/2 - n/4} = Z_{\text{SK}}(J)$ at $\beta = 1$, so the planted and ordinary laws at time s are EAMS’s $\mathbb{P}$ and $\mathbb{Q}$ at $t = s$, and the transfer display is (4.14), $\mathbb{E}_{\mathbb{P}}[F] = \mathbb{E}_{\mathbb{Q}}[Z_{\text{SK}}F]$, on $\{L \geq n^{-C}\}$.

What is new in the preprint.

- Work at $\beta = 1$, with explicit likelihood costs ((7.5), $\mathbb{E}L^2 \leq C(n + 1)$, $\log Z - \log \mathbb{E}Z = o(n)$) in place of contiguity, which it disclaims (§4).
- Stability on the whole approximate-root region, proved at $\beta = 1$ by Kac–Rice counts under the planted law (set-up credited to [26, 27]) with new shrinking-overlap estimates.
- Covariance and Dirichlet-form control down to the critical scale, matrix-precision observations, and the Haar-frame identity for $Z(Uc)/Z_o(Uc)$ (§5.1).

How to check.

1. *Observation.* §5.1, p. 12 against EAMS (1.4), p. 3, and (4.12), pp. 11–12.
2. *Planted law.* §5.1, p. 12 against EAMS (4.3)–(4.6), p. 10, and (4.9), p. 11; check that $\mathbb{E}Z(0) = e^{n/4}$ gives $Z(0)/\mathbb{E}Z(0) = Z_{\text{SK}}(J)$ at $\beta = 1$.
3. *Likelihood transfer.* The display in the proof of Proposition 14.1 (p. 107) and the transfer paragraph closing the proof of Lemma 17.3 (p. 145) against EAMS Proposition 4.2, p. 11, and (4.14), p. 12.
4. *Field equation.* §3, p. 7 and §4.1, p. 9 against EAMS (4.53), p. 20, via the identity above.

3.2 S2. The spherical saddle-point method with a field (Kosterlitz–Thouless–Jones, Baik et al., Landon, Du–Huang), in one- and two-replica form

Summary. The preprint’s spherical reference model is built on the Kosterlitz–Thouless–Jones (KTJ) saddle-point representation of the spherical partition function with a field [3, Lemma 3.1], in the Gaussian-conditioned-on-its-radius form evaluated at the mean radius [6, proof of Prop. 4.4]. A stronger form of its critical saddle location, with the same two-step proof, is [4, Prop. 3.1 and Lemma 3.4], and its normalization bound is a weak form of Landon’s Theorem 1.1. These are (5.4), (5.5) and Lemmas 5.3–5.4 in §5.3. Applied to two independent replicas, the same representation Fourier-inverts the joint law of their squared lengths and overlap ((7.2) and the paragraph after it, p. 27), as in [6, Lemma 4.3; proof of Prop. 4.5]; the Laplace (double-contour) form is [5, Lemma 4] and, with a field, [3, Lemma 3.3]. The method recurs in §6–§7, §11–§12, §15 and §19. The preprint cites none of KTJ, Baik–Lee, Baik et al., Landon or Nguyen–Sosoe, and cites the companion [24] only for other content. It cites [6, Thm. 1.6 and §3.3] at the head of §7 (p. 26) for the zero-field partition-function comparison, but that section uses their §4 only through its conclusion (their Prop. 3.1), and §4, which contains the representation, is not cited. **Grade: Strong**, because the saddle equation, the one- and two-replica representations and the shell formula agree term by term after renaming, and Landon proves a stronger saddle location by the same two steps.

Setup. Shared notation is in Section 2. For a field h , the preprint’s *saddle* is the unique $\lambda > \lambda_{\max}(J)$ with

$$D = (\lambda - J)^{-1}, \quad m_o = Dh, \quad \text{Tr } D + \|m_o\|^2 = n,$$

the preprint’s (5.4); as a function of λ the left side decreases strictly from ∞ to 0. The preprint states, without proof, that the spherical Gibbs law $\propto e^{x^\top Jx/2 + h^\top x} \pi_o(dx)$ is precisely $N(m_o, D)$ conditioned on $\|x\|^2 = n$. A superscript 0 marks zero field, so $\text{Tr } D^0 = n$. The eigenvalues of J are $\lambda_1 \geq \dots \geq \lambda_n$, with unit eigenvectors v_k ; put $\hat{h}_k = \langle v_k, h \rangle$ and $D_k = (\lambda - \lambda_k)^{-1}$. Write $f_{m,C}(t)$ for the density of $\|Y\|^2$ at t when $Y \sim N(m, C)$, and $f_C = f_{0,C}$. With $\ell = \log \log(n + 1000)$ and $d_0 = n^{-1/3}\ell$, (5.5) derives $|\lambda^0 - 2| \leq Cd_0^2$ and $\|D^0\| = (\lambda^0 - \lambda_{\max}(J))^{-1} \leq Cnd_0$ from the edge bounds (5.1) of Lemma 5.1 in two steps: at $2 + Cd_0^2$ the normalized trace deficit $1 - \frac{1}{n} \text{Tr}(2 + Cd_0^2 - J)^{-1}$ is of order d_0 , so $\lambda^0 < 2 + Cd_0^2$; at distance $c/(nd_0)$ above $\lambda_{\max}(J)$ the top pole alone closes that deficit.

For two replicas put $L = n^{2/3}$. Independent Gaussian replicas $x_a = m_o + D^{1/2}\eta_a$ ($a = 1, 2$, η_a white) have the Gram statistic $\mathbf{G} = ((x_a^\top x_b - n\delta_{ab})/L)_{a,b}$, the preprint’s (7.2), and characteristic function $\varphi(T) = \mathbb{E} e^{i \text{Tr}(T\mathbf{G})}$, $T \in \text{Sym}_2(\mathbb{R})$. Two independent spherical replicas are these conditioned on $\mathbf{G}_{11} = \mathbf{G}_{22} = 0$, so their overlap $x_1^\top x_2/L$ has density $f_{\mathbf{G}}(0, 0, u)/(f_{\mathbf{G}_{11}}(0)f_{\mathbf{G}_{22}}(0))$ at u .

Role in the argument. In Route 1 the sphere is the reference model for the cube (§5.1), and the representation puts the spherical posterior in closed form, with covariance at most $(1 + o(1))D$ (5.6). Lemma 5.4 locates the saddle under the observation field, $\lambda = 2 + (1 \pm \epsilon)s$ and $\|m_o\|^2 = (1 \pm 2\epsilon)nr$ up to probability $e^{-c\epsilon k}$, so $\|D\| \approx 1/s$, the target scale of Proposition 11.1; it is used in §11 and §12. Lemma 5.3 proves $e^{n/4}n^{-C} \leq Z_o(0) \leq e^{n/4+o(n)}$ from the shell formula below, bounding its density ratio by Lemma 5.2 (finding P2, Section 4.2). Proposition 7.1 compares cube and sphere at the edge: outside Haar mass $o(1)$, $Z(h)/Z_o(h) = 1 + o(1)$, and the second moments about m_o agree up to operator-norm errors. In its proof only the cube–cube term of a four-term signed identity for $H_U(x) = (x - m_o)(x - m_o)^\top - D$ (p. 27) is a lattice sum; comparing it with the integral needs bounds on the spherical pair-overlap density, with the insertion, and on its u -derivatives, which the two-replica representation supplies. Combined with Lemma 5.3,

Proposition 7.1 yields $Z(0) \geq n^{-C} e^{n/4}$ (7.5), which undoes the planted size bias; by itself it supplies the moment errors used for Proposition 12.3. The one-replica device recurs, uncited, in §6.1, in the proofs of Proposition 11.1 and Lemma 11.2, in the capped saddle defining R_p (15.3), in (15.14)–(15.15) and in (19.9); the two-replica device in the proofs of Lemma 11.4, Proposition 12.4 (several replicas) and Proposition 15.1 (an off-diagonal tilt).

The earlier technique. *The KTJ representation.* Fourier-inverting the constraint $\|\sigma\|^2 = N$ on a standard Gaussian turns the spherical partition function into one contour integral over the Lagrange multiplier z [28]. [2, Lemma 1.3, p. 5] state it at zero field and [3, Lemma 3.1, p. 11] with a field. There M is GOE with variances $(1 + \delta_{ij})/N$, eigenvalues $\lambda_1 \geq \dots \geq \lambda_N$ and eigenvectors u_i ; $\mathcal{H}(\sigma) = -\frac{1}{2}\sigma \cdot M\sigma - h g \cdot \sigma$ and $Z_N = \int e^{-\beta \mathcal{H}} d\omega_N$, with ω_N normalized on the sphere of radius $\sqrt{N}$ (eqs. (1.1)–(1.4)). For every $\gamma > \lambda_1$,

$$Z_N = C_N \int_{\gamma-i\infty}^{\gamma+i\infty} e^{\frac{N}{2}G(z)} dz, \quad C_N = \frac{\Gamma(N/2)}{2\pi i (N\beta/2)^{N/2-1}},$$

$$G(z) = \beta z - \frac{1}{N} \sum_{i=1}^N \log(z - \lambda_i) + \frac{h^2 \beta}{N} \sum_{i=1}^N \frac{n_i^2}{z - \lambda_i}, \quad n_i = u_i \cdot g.$$

By their eq. (5.3) (p. 17), G' increases from $-\infty$ to β on (λ_1, ∞) , so a unique saddle $\gamma > \lambda_1$ solves $G'(\gamma) = 0$.

The density-at-the-mean form. Instead of steepest descent through γ (Baik–Lee, Landon), one can stop at the saddle: the Gibbs law is the Gaussian $N(0, (\gamma - M)^{-1})$ conditioned on its squared radius, and Z is a Gaussian normalization times that radius's density at N , which at the saddle is its mean. Du and Huang use the preprint's normalization, with their W (not the planted noise) in place of J . In [29, §1.2, eq. (1.7), pp. 4–5], γ solves $\sum_i (\gamma - \lambda_i)^{-1} = N$ and $\mu_{N,1}^{\text{sph}}$ is precisely the law of $\xi \sim N(0, (\gamma I - W)^{-1})$ conditioned on $\|\xi\|^2 = N$. In the proof of [6, Prop. 4.4, p. 22], from the KTJ formula (their Lemma 4.2, p. 18, with $C_N = \Gamma(N/2)/(N/2)^{N/2-1}$), they set $a_k = \gamma - \lambda_k$ and $\Xi_W = \sum_k \xi_k^2/a_k$ with ξ_k i.i.d. $N(0, 1)$, so $\mathbb{E} \Xi_W = N$, and obtain the identity displayed below, where ρ_W is the density of Ξ_W .

Landon at criticality. [4, eqs. (2.3)–(2.5), pp. 4–5] takes H GOE with variances $1/N$ and $2/N$ and normalizes $Z_N(\beta)$ on the sphere of radius $\sqrt{N}$, so $Z_N(1)$ is $Z_o(0)$ with $J = H$; the saddle solves $\beta = \frac{1}{N} \sum_i (\gamma - \lambda_i)^{-1}$ (eq. (3.2), p. 9). At $\beta = 1$, Prop. 3.1 (eq. (3.3), p. 9) and Lemma 3.4 (p. 10) put both $N^{2/3}(\gamma - \lambda_1)$ and $N^{2/3}(\gamma - 2)$ in $[C_\varepsilon^{-1}, C_\varepsilon]$ with probability at least $1 - \varepsilon$, in two steps. By eq. (3.7), above the edge $\frac{1}{N} \sum_j (E - \lambda_j)^{-1}$ falls below 1 by at least $cN^{-1/3} \sqrt{N^{2/3}(E - 2)} - N^{-1/3}$, which bounds γ from above; the top-pole term $N^{-1/3}/(N^{2/3}(E - \lambda_1))$ of the decomposition (3.5) (Lemma 3.2) closes this deficit near λ_1 , which bounds $\gamma - \lambda_1$ from below. Thm. 1.1 (eq. (1.6), p. 2) is the critical central limit theorem $(\log Z_N(1) - \frac{N}{4} + \frac{1}{12} \log N)/\sqrt{\frac{1}{6} \log N} \Rightarrow N(0, 1)$, where $f(1) = \frac{1}{4}$ is the limiting free energy (eq. (1.3), p. 1), computed at zero field by KTJ [3, p. 4].

Two replicas. In Laplace form, for spherical SK at zero field (any $\beta > 0$), [5, Lemma 4, eqs. (3.2)–(3.4), p. 6] write $\langle e^{tR_{12}} \rangle$, normalized by the square of the one-replica integral, as a double integral over $z, w \in \gamma + i\mathbb{R}$ with kernel $(z - \lambda_i)(w - \lambda_i) - t^2/(4\beta^2 N^2)$. [3, Lemma 3.3, eqs. (3.5)–(3.6), p. 12; App. A, pp. 70–71] add the field: $\langle e^{\eta \Re} \rangle$ is the ratio of $\iint e^{\frac{N}{2} \mathcal{G}_{\Re}(z, w; a)}$ at $a = \eta/(\beta N)$ and at $a = 0$, where $\mathcal{G}_{\Re}$ has kernel $(z - \lambda_i)(w - \lambda_i) - a^2$ and field term $\frac{h^2 \beta}{N} \sum_i n_i^2 (z + w - 2\lambda_i + 2a)/((z - \lambda_i)(w - \lambda_i) - a^2)$. In Fourier/density form, at $\beta = 1$ and $h = 0$, [6, Lemma 4.3, p. 19] write the partition function $Z_{N,2}^{\text{sph}}$ of two orthogonal spherical replicas as an integral over $(\gamma, \gamma, 0) + (i\mathbb{R})^3$ with kernel $\det(Z - \lambda_k I_2)$, $Z = \begin{pmatrix} z_1 & w \\ w & z_2 \end{pmatrix}$. In the proof of their Prop. 4.5 (pp. 23–24), with (g_k, h_k) i.i.d. $N(0, I_2)$,

$T_1 = \sum_k g_k^2/a_k$, $T_2 = \sum_k h_k^2/a_k$ and $T_3 = 2 \sum_k g_k h_k/a_k$, they compute $\mathbb{E} e^{-\frac{i}{2}(t_1 T_1 + t_2 T_2 + s T_3)}$ as a product of 2×2 determinants (below) and invert it: $Z_{N,2}^{\text{sph}} = C_{N,2} e^{N\gamma - \log \det(\gamma - W)} \cdot 8\rho_{W,2}(N, N, 0)$, with $\rho_{W,2}$ the density of (T_1, T_2, T_3) .

Correspondence. At $\beta = 1$ and $N = n$; at $h = 0$ also $\lambda^0 = \gamma$ and $D_k^0 = 1/a_k$:

Preprint	Earlier work
J (GOE, variances $1/n, 2/n$) (§5.1); field h ; $Z_o(h)$	M, hg, Z_N (Baik et al., eqs. (1.1), (1.4), p. 2; GOE on p. 3); $H, Z_N(1)$ (Landon, eqs. (2.3), (2.5)); W, Z_N^{sph} (Du–Huang)
saddle λ ; $D = (\lambda - J)^{-1}$ (5.4)	saddle γ ; $\sum_i (\gamma - \lambda_i)^{-1} = \text{Tr}(\gamma - M)^{-1}$ (Baik et al., eq. (5.3))
Gibbs law is $N(m_o, D)$ conditioned on $\ x\ ^2 = n$ (§5.3, p. 17)	zero field: Du–Huang, overlap paper, eq. (1.7)
$f_{D^0}(n)$; $f_I(n)$	$\rho_W(N)$; the χ_N^2 density at N , implicit in C_N (Du–Huang, Lemma 4.2, p. 18; $2C_N = e^{-N/2}/f_I(N)$)
saddle location (5.5), proved from (5.1) (p. 17)	Landon, Prop. 3.1 and Lemma 3.4, proved from (3.5) and (3.7)
Lemma 5.3 (5.7); its constant $\frac{1}{4}$ (p. 19)	Landon, Thm. 1.1; $f(1) = \frac{1}{4}$ (eq. (1.3))
Gaussian replicas, G ((7.2), p. 27); overlap density at u	$(T_1, T_2, T_3) = (n + LG_{11}, n + LG_{22}, 2LG_{12})$ at $h = 0$; $Z_{N,2}^{\text{sph}}/(Z_N^{\text{sph}})^2 \propto \rho_{W,2}(N, N, 0)/\rho_W(N)^2$ (Du–Huang, pp. 22–24)
frequency $T \in \text{Sym}_2(\mathbb{R})$; noncentral factor (p. 27)	$\Theta = -2T/L$ (Du–Huang, proof of Prop. 4.5); field term of Baik et al. (3.5)
off-diagonal tilt (proofs of Lemma 11.4 and Proposition 15.1)	Nguyen–Sosoe, Lemma 4, (3.8)–(3.10); Baik et al., Lemma 3.3

The key formulas:

Preprint	Earlier work
Saddle (5.4): $\text{Tr}(\lambda - J)^{-1} + \ (\lambda - J)^{-1}h\ ^2 = n$	Baik et al., $N G'(\gamma) = 0$ at $\beta = 1$, field hg : $\sum_i \frac{1}{\gamma - \lambda_i} + \sum_i \frac{(hn_i)^2}{(\gamma - \lambda_i)^2} = N$
Shell formula (p. 19): $\log Z_o(0) = \frac{n(\lambda^0 - 1)}{2} - \frac{1}{2} \log \det(\lambda^0 - J) + \log \frac{f_{D^0}(n)}{f_I(n)}$	Du–Huang, proof of Prop. 4.4: $Z_N^{\text{sph}} = C_N e^{\frac{N\gamma}{2} - \frac{1}{2} \log \det(\gamma - W)} \cdot 2\rho_W(N)$

With a field, the same computation gives, for every $\lambda > \lambda_{\max}(J)$,

$$\log Z_o(h) = \frac{n(\lambda-1)}{2} - \frac{1}{2} \log \det(\lambda - J) + \frac{1}{2} h^\top D h + \log \frac{f_{Dh, D}(n)}{f_I(n)},$$

and Fourier inversion of the noncentral χ^2 characteristic function turns it into Baik et al.'s contour integral on $\Re z = \lambda$. The preprint's (15.14)–(15.15) is this identity with J_p for J and $\lambda = 2 + \alpha$.

For two replicas at $h = 0$, put $\Theta = \begin{pmatrix} t_1 & s \\ s & t_2 \end{pmatrix} = -2T/L$. The preprint's (undisplayed) transform (p. 27) equals Du–Huang's integrand $e^{iN(t_1+t_2)/2} \mathbb{E} e^{-\frac{i}{2}(t_1 T_1 + t_2 T_2 + s T_3)}$:

$$\varphi(T) = e^{-in \operatorname{Tr} T/L} \prod_k \det(I_2 - \frac{2i}{L} D_k^0 T)^{-1/2} = e^{iN(t_1+t_2)/2} \prod_k \det(I_2 + \frac{i}{a_k} \Theta)^{-1/2},$$

so its Fourier inversion is theirs, evaluated at $(0, 0, u)$ instead of $(0, 0, 0)$; with $Z = \lambda I_2 - 2iT/L$ each factor is $(\lambda - \lambda_k) \det(Z - \lambda_k I_2)^{-1/2}$, the Lemma 4.3 kernel. A field multiplies $\varphi(T)$ by the noncentral factor $e^{\Psi(Z) - \Psi(\lambda I_2)}$, where $\Psi(Z) = \frac{1}{2} \sum_k \hat{h}_k^2 \mathbf{1}^\top (Z - \lambda_k I_2)^{-1} \mathbf{1}$. For $Z = \begin{pmatrix} z & -a \\ -a & w \end{pmatrix}$ and $\hat{h}_i = h n_i$, $\mathbf{1}^\top (Z - \lambda_i I_2)^{-1} \mathbf{1} = (z + w - 2\lambda_i + 2a) / \det(Z - \lambda_i I_2)$, so $\Psi(Z)$ is $\frac{N}{2}$ times Baik et al.'s field term at $\beta = 1$.

What is new in the preprint.

- The applications: the saddle under the planted observation field at criticality, with tails at scale $k = nr^3$ (Lemma 5.4), and the capped and frozen versions in §15–§19.
- The edge inputs: tridiagonal-model bounds (Lemma 5.1) replace rigidity and a log-determinant central limit theorem. Against Landon's results, (5.5) loses powers of ℓ , and (5.7) loses a polynomial factor below and $e^{o(n)}$ above.
- In the two-replica form: a general overlap u with fixed-order u -derivatives, at fixed spectrum (Du–Huang need only $(N, N, 0)$, and their kernel bounds average over the GOE using log-concavity, [6, Lemma 3.3, Cor. 3.4]); a field in the Fourier/density form; and the insertions H_U , with a Wick analysis under the complex tilt $\Delta_T = (I - 2iT \otimes D/L)^{-1} - I$ ((7.3)–(7.4), p. 28).

How to check.

1. Compare the saddle and representation (5.4) and the sentence after it with [3, Lemma 3.1, p. 11; eq. (5.3), p. 17] at $\beta = 1$ with $hg = h$, and with [29, eq. (1.7), pp. 4–5].
2. Compare the shell formula (proof of Lemma 5.3, p. 19) with [6, Lemma 4.2, p. 18; proof of Prop. 4.4, p. 22]: take logarithms and use $2C_N = e^{-N/2}/f_I(N)$. Then compare (15.15) with Baik et al.'s Lemma 3.1 at $\gamma = 2 + \alpha$ and $M = J_p$, noting that the preprint's constant $C_{n,\alpha} = \log(2^{n/2} \Gamma(n/2) n^{1-n/2}) + \frac{n}{2}(2 + \alpha)$ is $\frac{n(\lambda-1)}{2} - \log f_I(n)$.
3. Compare the saddle location and its proof (5.5), p. 17 with [4, Prop. 3.1 with (3.5) and (3.7), p. 9; Lemma 3.4, p. 10]: the deficit at $2 + Cd_0^2$ against (3.7), and the top pole at distance $c/(nd_0)$ against (3.5). Since $C_\varepsilon \leq \ell$ eventually, Landon's bounds imply (5.5).
4. Compare the normalization (5.7) with [4, Thm. 1.1, p. 2; eqs. (1.3), (2.3)–(2.5)]; it implies (5.7) for any $C > 1/12$.
5. Compare (7.2), the paragraph after it (p. 27) and Δ_T after (7.3) (p. 28) with [6, proof of Prop. 4.5, pp. 23–24], and the noncentral-factor sentence (p. 27) with [3, eq. (3.5), p. 12; App. A, p. 71].
6. Compare the proofs of Lemma 11.4 (p. 64) and Proposition 12.4 (p. 81) with [6, Lemma 4.3, p. 19] and [5, eqs. (3.8)–(3.10), p. 7]: there $\det(I - 2iQ)$, with Q the frequency tensored with the covariance over L , factors by mode into the kernel $(z - \lambda_i)(w - \lambda_i) - a^2$.

3.3 S3. The Brézin–Hikami duality for GOE characteristic polynomials with an external source

Summary. The polynomial bound on the second moment of the Kac–Rice Jacobian determinant (Lemma 8.2, pp. 31–32) rests on an exact identity (p. 32) that writes $\mathbb{E} \det(\cdot)^2$, for a GOE matrix with a diagonal shift and variance profile, as a six-variable Gaussian average of a product of per-site factors. It is the $k = 2$ case of Brézin and Hikami’s dual Grassmann representation of GOE characteristic polynomials with an external source [7, §6, eqs. (88)–(89), pp. 19–20, with (43), p. 10]: factoring out the variance profile turns it into their diagonal source, and the preprint’s derivation follows their §3 step by step. The preprint introduces the identity with “The exact identity is” and derives it “for completeness”, citing no source; the opening of §8 (p. 30) credits only the Kac–Rice organization and lists the “polynomial determinant bound” among results “proved here”. Brézin–Hikami appears nowhere in the preprint or its companions. Grade: STRONG, because the identity and its derivation are reused exactly and carry that bound.

Setup. For a candidate root y of the TAP-type equation of Section 2, put $b = 1 - q$, $v = 1 - m^2 \in (0, 1]^n$ and $V = \text{diag } v$ (§8, p. 30), and let $\mathbf{Q}_y$ be the law of J conditioned on y being a root. The Kac–Rice formula (Lemma 8.1, p. 31) integrates over y an explicit scalar root density times $\mathbb{E}_{\mathbf{Q}_y}[T\mathcal{W}]$, for any nonnegative test $T(y, J)$, where

$$\mathcal{W}(y, J) = e^{-nb^2/2} |\det(I + (bI - J - 2mm^\top/n)V)| \quad (8.1)$$

is the normalized absolute Jacobian. Lemma 8.2 states $\mathbb{E}_{\mathbf{Q}_y} \mathcal{W}^2 \leq n^{C_D}$ on $\|y\| \leq n^D$. For an unconditioned GOE matrix W_0 , real z and $x_i = 1 + z + bv_i$, the identity in its proof (p. 32) is

$$\mathbb{E}_{W_0} \det((1+z)I + bV - \sqrt{V}W_0\sqrt{V})^2 = \mathbb{E}_H \prod_{i=1}^n \left[(x_i + iv_i H_1)(x_i + iv_i H_2) + v_i^2(|h_1|^2 + |h_2|^2) \right], \quad (\text{P})$$

where H_1, H_2 are real $N(0, 1/n)$ and h_1, h_2 are circular complex Gaussians with $\mathbb{E}|h_j|^2 = 1/n$, all independent. The bracketed terms are the *site factors*. The derivation (p. 32) uses exterior (Grassmann) variables $\bar{\psi}_i^a, \psi_i^a$, $a = 1, 2$: averaging W_0 adds $-(X_{11}^2 + X_{22}^2 + 2X_{12}X_{21} + 2PQ)/(2n)$ to the exponent, with local symbols $X_{ab} = \sum_i v_i \bar{\psi}_i^a \psi_i^b$, $P = \sum_i v_i \bar{\psi}_i^1 \bar{\psi}_i^2$ and $Q = \sum_i v_i \psi_i^1 \psi_i^2$, and these terms are decoupled with H_1, H_2, h_1, h_2 .

Role in the argument. On Route 1 (Section 2), Sections 8–10 show that, under the planted law, every root at the observed field passes scalar and matrix diagnostics, which feed the §11 comparison. Each diagnostic is first proved without the weight $\mathcal{W}$, with failure probability $n^C e^{-Bk}$, and then transferred to the root count by $\mathbb{E}_{\mathbf{Q}_y}[\mathbf{1}_F \mathcal{W}] \leq \mathbf{Q}_y(F)^{1/2} (\mathbb{E}_{\mathbf{Q}_y} \mathcal{W}^2)^{1/2}$. Polynomial accuracy is essential, since $k = nr^3$ need only exceed $\log^{10} n$ (§8, p. 30): a loss of $e^{\varepsilon n}$, as in the Kac–Rice works credited for the set-up [26, 27], would swamp e^{-Bk} . In the proof, symmetrization and a fixed-rank comparison of J under $\mathbf{Q}_y$ with W_0 reduce the bound to the left side of (P), uniformly in real $|z| \leq 1/n$ (pp. 31–32); then (P), a contour shift and an elementary estimate give $\leq Cn^{3/2}$ (p. 32). The lemma is invoked at Proposition 8.3 (p. 33), §10.1 (p. 42) and, implicitly, Proposition 9.1 (p. 37).

The earlier technique. Brézin and Hikami [7] (arXiv version; printed page numbers) compute $F_k(\lambda_1, \dots, \lambda_k) = \mathbb{E} \prod_{\alpha=1}^k \det(\lambda_\alpha - \mathbf{X})$ for a real symmetric $N \times N$ matrix $\mathbf{X}$ with weight $e^{-\frac{N}{2} \text{Tr } \mathbf{X}^2}$ ((21), p. 6), so $\mathbb{E} \mathbf{X}_{ii}^2 = 1/N$ and $\mathbb{E} \mathbf{X}_{ij}^2 = 1/(2N)$, by trading the $N \times N$ average for an

integral over $k \times k$ matrices. Each determinant is a Gaussian Grassmann integral, $\det(\lambda - X) = \int \prod_a d\bar{c}_a dc_a e^{iN\bar{c}_a(\lambda\delta_{ab} - X_{ab})c_b}$ ((6)–(7), p. 3), with one copy $c_{a\alpha}$ per factor α . The GOE average is then Gaussian, $\mathbb{E}e^{iN\text{Tr}XY} = e^{-\frac{N}{4}\text{Tr}(Y^2 + YY^\top)}$ with $Y_{ab} = -\sum_\alpha \bar{c}_{a\alpha}c_{b\alpha}$ ((22), p. 6), and yields two quartic terms: $\text{Tr}Y^2 = -\text{tr}\gamma^2$ with $\gamma_{\alpha\beta} = \sum_a \bar{c}_{a\alpha}c_{a\beta}$ ((10), (23)), already present for the GUE, and the pairing term $\text{Tr}YY^\top = \text{tr}UV$ with $U_{\alpha\beta} = \sum_a \bar{c}_{a\alpha}\bar{c}_{a\beta}$, $V_{\alpha\beta} = \sum_a c_{a\alpha}c_{a\beta}$ ((24)–(26), pp. 6–7), which is specific to orthogonal symmetry. A Hubbard–Stratonovich transform decouples the first with a $k \times k$ Hermitian B ((11), p. 4) and the second with a complex antisymmetric D ((27), p. 7). The exponent is then a sum over sites, so the Grassmann integral factorizes into one Pfaffian per site ((28)–(30), p. 7):

$$F_k = \int dB dD e^{-N\text{tr}(B^2 + D^\dagger D)} [-\text{Pf } \mathbf{M}]^N, \quad \mathbf{M} = \begin{pmatrix} D & \Lambda - iB^\top \\ -(\Lambda - iB) & D^\dagger \end{pmatrix} \quad ((31)–(32)),$$

with $\Lambda = \text{diag}(\lambda_\alpha)$ and $2k^2 - k$ real integration variables (pp. 7–8). For $k = 2$, D has a single entry d and

$$-\text{Pf } \mathbf{M} = |d|^2 + (\lambda_1 - iB_{11})(\lambda_2 - iB_{22}) + |B_{12}|^2 \quad ((42)–(43), \text{ pp. 9–10}),$$

the quaternion determinant of a 2×2 self-dual quaternion matrix: the GOE $\rightarrow$ GSE duality (47). In §6 (pp. 19–20) a diagonal source $A = \text{diag}(a_j)$ is added ((86), p. 19), and, following (88), F_k is read for $\mathbb{E} \prod_\alpha \det(\lambda_\alpha + A - X)$ ((86) as printed gives $\lambda_\alpha - A$; the sign of a diagonal source is immaterial here). Being diagonal, the source enters only the Pfaffian of site j , through $\Lambda \mapsto \Lambda + a_j \mathbf{1}$ in $\mathbf{M}^{(j)}$ ((88), p. 19), so

$$F_k = \int dB dD e^{-N\text{tr}(B^2 + D^\dagger D)} \prod_{j=1}^N [-\text{Pf } \mathbf{M}^{(j)}] \quad ((89), \text{ p. 20}).$$

Correspondence. Throughout, $n = N$, $k = 2$, and $W_0 \stackrel{d}{=} \sqrt{2}X$.

Preprint (proof of Lemma 8.2, p. 32)	Brézin–Hikami [7]
$W_0, \mathbb{E}W_{ij}^2 = (1 + \delta_{ij})/n$ (§2, p. 6)	X , weight $e^{-\frac{N}{2}\text{Tr}X^2}$ ((21), p. 6)
$\mathbb{E}\det(\Gamma_z - W_0)^2, \Gamma_z = \text{diag}(x_i/v_i)$	$2^n F_2(\lambda, \lambda)$, source $\lambda + a_j$ ((86), (88), p. 19)
$\bar{\psi}_i^a, \psi_i^a$: site i , flavour $a = 1, 2$	$\bar{c}_{a\alpha}, c_{a\alpha}$: site a , flavour $\alpha = 1, 2$ ((6)–(7), p. 3)
$X_{11}^2 + X_{22}^2 + 2X_{12}X_{21}$	$\text{tr}\gamma^2 = -\text{Tr}Y^2$ ((10), (23))
$2PQ$	$\text{Tr}YY^\top = \text{tr}UV$ ((24)–(26))
H_1, H_2 real, h_1, h_2 complex: six real variables	Hermitian B , antisymmetric D : $2k^2 - k = 6$ ((11), (27))
$\mathbb{E}_H \prod_i [\text{site factor}]$	(89), site factor $-\text{Pf } \mathbf{M}^{(j)}$ ((43), (88))

Since every $v_i > 0$, $(1+z)I + bV - \sqrt{V}W_0\sqrt{V} = \sqrt{V}(\Gamma_z - W_0)\sqrt{V}$ with $\Gamma_z = \text{diag}((1+z)/v_i + b)$, so the variance profile is a diagonal source and the left side of (P) is $2^n(\det V)^2 F_2(\lambda, \lambda)$ with $\lambda = b/\sqrt{2}$ and $a_j = (1+z)/(\sqrt{2}v_j)$, so that $\lambda + a_j = x_j/(\sqrt{2}v_j)$. Under $H_l = -\sqrt{2}B_{ll}$, $h_1 = \sqrt{2}B_{12}$,

$h_2 = \sqrt{2}d$, the weight $e^{-N(B_{11}^2+B_{22}^2+2|B_{12}|^2+2|d|^2)}$ of (89) becomes the law of (H_1, H_2, h_1, h_2) in (P), and the site factors match:

$$\underbrace{(x_j + iv_j H_1)(x_j + iv_j H_2) + v_j^2(|h_1|^2 + |h_2|^2)}_{\text{preprint, (P)}} = 2v_j^2 \underbrace{[|d|^2 + (\lambda + a_j - iB_{11})(\lambda + a_j - iB_{22}) + |B_{12}|^2]}_{-\text{Pf } M^{(j)}, \text{ BH (43), (88)}}$$

The factors $\prod_j 2v_j^2$ are those on the left, so (P) is (89) at $k = 2$ and $\lambda_1 = \lambda_2$.

What is new in the preprint.

- The contour shift $H_l \mapsto H_l + ib$, $l = 1, 2$, whose cost e^{nb^2} cancels exactly the normalization e^{-nb^2} in $\mathcal{W}^2$.
- The finite- n polynomial bound: the modulus bound through e_1, e_2 , the estimate $e^{-cn \min(R_H^4, R_H^2)}$, in which the quadratic terms cancel exactly, and the integral bound $\leq Cn^{3/2}$, uniformly in $z \in [-1/n, 1/n]$ and $v \in (0, 1]^n$. At $v \equiv 1$, $z = 0$ the source is $a_j = 1/\sqrt{2}$, which is Brézin and Hikami's critical value c [7, (91), p. 20]; they reach a quartic action there only by formal large- N asymptotics.
- The reduction from $\mathbf{Q}_y$ to an unconditioned GOE, and the use of the bound inside a shrinking-overlap Kac–Rice argument.

How to check.

1. Compare the identity in the proof of Lemma 8.2, p. 32 with [7, eqs. (88)–(89), pp. 19–20, and (43), p. 10]. Factor $\sqrt{V}$ out of the determinant, set $k = 2$ and $\lambda_1 = \lambda_2$, and apply the dictionary above; the site factors and Gaussian weights coincide.
2. Compare the derivation on p. 32 with [7, eqs. (6)–(7), p. 3, and (22)–(30), pp. 6–7]: the terms $X_{11}^2 + X_{22}^2 + 2X_{12}X_{21}$ and $2PQ$ are $\text{tr } \gamma^2$ and $\text{tr } UV$, decoupled by a 2×2 Hermitian matrix and one antisymmetric entry.
3. Read p. 32 and the opening of §8 (p. 30): neither cites Brézin–Hikami, and the latter lists the polynomial determinant bound among results “proved here”.

3.4 S4. Polchinski renormalization: trickle-down of Hessian bounds across caps

Summary. The cap induction of the preprint carries a Hessian bound for the capped potential ϕ_p from each spectral cap to the next finer one. Each cap step is one discrete step of the Polchinski renormalization scheme of Bauerschmidt–Bodineau [8], in the form set out in the survey of Bauerschmidt–Bodineau–Dagallier (BBD) [9, §§3.2, 3.4–3.5], and the steps compose across caps by the semigroup property. In stochastic-localization language it is the trickle-down covariance induction of Anari–Koehler–Vuong (AKV) [10, §3], and its scalar bound is their trickle-down of semi-log-concavity [10, Thm. 102]. The step occupies §§16.1–16.2: the convolution and transition (16.3)–(16.4), the Hessian identity (16.11) and the Brascamp–Lieb bound (16.14)–(16.15), each matching a numbered equation of BBD with a sign flip ((16.14)–(16.15) after recentring); it is iterated in the proof of Theorem 18.6 and reused in §18. The preprint cites none of the sine-Gordon paper, BBD or Bauerschmidt–Dagallier; it cites AKV and an earlier Bauerschmidt–Bodineau paper only for their SK temperature thresholds, and near the identities it credits only the Brascamp–Lieb inequality and, for the form of a two-sided covariance bound, Chewi–Pooladian [30] (§16.2, p. 129). **Grade:** STRONG, because the whole one-step apparatus (renormalized potential, fluctuation measure, Hessian identity, Brascamp–Lieb bound under a margin) is reused intact and composed across the caps by the semigroup property.

Setup. Section 2 defines $d, P_p, k_p, B_p, J_p, \phi_p, R_p$ and α . In addition:

- *Cap and Hessian.* Put $H_p = \text{ran } P_p$, with fields $h \in H_p$. The capped Gibbs law $\mu_{p,h} \propto e^{x^\top J_p x / 2 + h^\top x}$ is the posterior of $X \sim \mu_J$ after a Gaussian observation of precision B_p with value h (§2), so a larger p means more observation and a flatter interaction. The projected Hessian is $K_p = \nabla_{H_p}^2(\phi_p|_{H_p}) = \text{Cov}_{\mu_{p,h}}(P_p X)$ (§16, p. 124).
- *Invariant.* With $R_p^H = (p^2 + \alpha_p(h))^{-1} I_{H_p}$, $e = h/\|h\|$ and $D = p^2(K_p - R_p^H)$, the induction carries (16.2)

$$\|D_{\perp\perp}\| \leq \delta, \quad \|D_{\perp e}\| \leq K_0 \sqrt{\delta}, \quad -K_0 I_{H_p} \preceq D, \quad D_{ee} \leq -0.30$$

on the annulus $\mathcal{A}_p(\sigma) = \{h \in H_p : |\alpha_p(h)| \leq \sigma p^2\}$ (16.1). The negative entry D_{ee} is the radial deficit.

- *One step.* For a child scale $c = xp$ (with $x = \gamma$ in the induction) put $\Delta = B_p - B_c \succeq 0$, $E = \text{ran } \Delta$ and $U = H_c \subset E$. Then $J_c = J_p + \Delta$ and $\Delta_U = (1 - x^2)p^2 I_U$; inverses of Δ are taken on E (§16.1, p. 126).
- *Auxiliary steps.* For a fixed number of initial levels ϕ_p is an auxiliary potential (Proposition 18.2), and the function convolved is a semiconcave extension $\bar{\phi}_p$, an infimum of supporting paraboloids of curvature $L_p = (1 + \gamma^4/100)p^{-2}I$ (16.7), with margin $\|\Delta_E^{1/2} L_p \Delta_E^{1/2}\| < 1$ (§16.1, p. 127).
- *Recentring.* With the child saddle parameter α frozen, $D^\circ = p^2(K_p - (p^2 + \alpha)^{-1}I)$, $C_0 = (\Delta_E^{-1} - R_p|_E)^{-1}$, $S = p^{-2}C_0$ and $N = (p^2 + \alpha)/(c^2 + \alpha)$ (§16.2, pp. 129–130).

Role in the argument. This is Route 2 of Section 2. Proposition 16.1 (“One cap step”) propagates the invariant from p to $c = \gamma p$, outside an event of probability at most $Ce^{-ak_p} + Ce^{-a'k_c}$ over a fresh Haar refinement. The step is iterated from the fixed initialization scale p_0 of §§17–18 down to $np^3 \asymp \sqrt{\log n}$. The output, Theorem 18.6, is $P_p \nabla^2 \phi_p(h) P_p \preceq (1 + \epsilon)p^{-2}P_p$ on a thin saddle annulus (and $\preceq Cp^{-2}P_p$ on bounded field balls). These curvature bounds control the information revealed along the forward observation path in §19, whose entropy-retention and layer argument gives the log-Sobolev inequality (Theorem 19.3). The same device recurs in §18, where the auxiliary potential is one convolution step and (18.13) is the Brascamp–Lieb bound under the margin (18.11); in single-step form in §19.4 (pp. 178–179); and, closed by a Poincaré estimate instead of Brascamp–Lieb, as the backward covariance induction of §12.3 on the isotropic route.

The earlier technique. Bauerschmidt–Bodineau’s sine-Gordon paper [8] (not their earlier JFA paper [31], which the preprint cites and which makes a single such step) splits a Gaussian covariance as $C_t = \int_0^t \dot{C}_s ds$ and integrates the fluctuations out scale by scale. For a potential V_0 , the renormalised potential and the renormalisation semigroup are [8, (1.3)–(1.4), p. 2]; [9, Def. 3.2, (3.11)–(3.12), p. 16]

$$e^{-V_t(\varphi)} = \mathbb{E}_{C_t}(e^{-V_0(\varphi+\zeta)}), \quad \mathbf{P}_{s,t}F(\varphi) = e^{V_t(\varphi)} \mathbb{E}_{C_t-C_s}(e^{-V_s(\varphi+\zeta)}F(\varphi+\zeta)),$$

and, by the semigroup property, from scale s to t , $V_t(\varphi) = -\log \mathbb{E}_{C_t-C_s} e^{-V_s(\varphi+\zeta)}$ [9, proof of Lemma 3.11, p. 22]. The kernel of $\mathbf{P}_{0,t}$ is the fluctuation measure $\mu_t^\varphi(d\zeta) \propto e^{-V_0(\zeta)} \mathbf{N}(\varphi, C_t)(d\zeta)$ [9, (3.17)–(3.18), p. 17]. Differentiating gives $\nabla V_t(\varphi) = \mathbb{E}_{\mu_t^\varphi}[C_t^{-1}(\varphi - \zeta)]$ and [9, Lemma 3.12, (3.60)–(3.61), p. 23]

$$(f, \text{Hess } V_t(\varphi)f) = \mathbb{E}_{\mu_t^\varphi}[(f, \text{Hess } V_0(\zeta)f)] - \text{Var}_{\mu_t^\varphi}((f, \nabla V_0(\zeta))) = (f, C_t^{-1}f) - \text{Var}_{\mu_t^\varphi}((C_t^{-1}f, \zeta)).$$

Bounding the variance by Brascamp–Lieb gives [9, Prop. 3.13, Proof 1, (3.63)–(3.65), p. 23]

$$\text{Hess } V_t(\varphi) \geq \mathbb{E}_{\mu_t^\varphi} [\text{Hess } V_0 (C_t^{-1} + \text{Hess } V_0)^{-1} C_t^{-1}], \quad C_t^{1/2} \text{Hess } V_t C_t^{1/2} \geq \mathbb{E}_{\mu_t^\varphi} \left[\frac{\hat{H}_t}{\text{id} + \hat{H}_t} \right],$$

with $\hat{H}_t = C_t^{1/2} \text{Hess } V_0 C_t^{1/2}$, in one step from 0 to t for convex V_0 . For non-convex V_0 in one dimension the step needs a margin, $t < t_0$ with $t_0^{-1} = -\inf V_0''$ [9, Ex. 3.15, p. 26]. Along the continuous flow the Hessian obeys [9, Lemma 3.11, (3.56)], and Bauerschmidt–Bodineau’s criterion also admits discrete decompositions $C_0 + C_1 + \dots$ [8, Thm. 2.6]. Bauerschmidt–Bodineau feed these bounds into a multiscale Bakry–Émery criterion [8, Thms. 1.2 and 2.5]. For Ising models the fluctuation measure is an Ising measure with spin covariance Σ_t , and $\text{Hess } V_t = C_t^{-1} - C_t^{-1} \Sigma_t C_t^{-1}$ [33, (1.23)]; BBD show that both that criterion and Chen–Eldan’s entropic stability then reduce to a bound on Σ_t uniform over external fields [9, Thm. 6.12, Ex. 6.15]. Bauerschmidt–Dagallier use this to get a polynomial log-Sobolev inequality for near-critical Ising models.

AKV run the same induction in stochastic-localization language, which BBD identify with the Polchinski flow [9, §4.5]. For a linear-tilt localization (ν_t) , [10, eq. (8), p. 32]

$$\text{Cov}(\nu_t) = \mathbb{E}[\text{Cov}(\nu_{t+1}) \mid \mathcal{F}_t] + \text{Cov}(\nu_t) \text{Cov}(Z_{t+1} \mid \mathcal{F}_t) \text{Cov}(\nu_t),$$

which carries covariance bounds backward from the more localized ν_{t+1} to ν_t ; this is the “trickle-down”. Their scalar trickle-down of semi-log-concavity [10, Thm. 102, p. 76], which they prove from its continuous-time form [10, (10)] rather than by Brascamp–Lieb, reads: if every tilt of μ has covariance $\preceq \gamma^{-1}I$ and $T < \gamma$, then every tilt of $\mathcal{G}_{-T}\mu$, where $d\mathcal{G}_{-T}\mu/d\mu \propto e^{T\|x\|^2/2}$, has covariance $\preceq (\gamma - T)^{-1}I$. Chen–Eldan’s Lemma 50 [32] is the special case of a product base measure on the cube, proved by one Gaussian decomposition step plus Brascamp–Lieb and credited there to [31].

Correspondence. The dictionary applies BBD’s Definition 3.2 to $V_s := -\phi_p$ (or $-\bar{\phi}_p$), with covariance increment Δ ; upper bounds on K become lower bounds on $\text{Hess } V$.

Preprint	Bauerschmidt–Bodineau, BBD, AKV
parent scale p , child scale $c = \gamma p < p$	times $s < t$ (BBD Def. 3.2)
$\Delta = B_p - B_c$ (§16.1)	$C_t - C_s$
child field h_c ; parent field $Y = h_p$	$\varphi; \zeta$ after the shift in BBD (3.18)
ϕ_p, ϕ_c (or $\bar{\phi}_p$); K_p, K_c	$-V_s, -V_t; -\text{Hess } V_s, -\text{Hess } V_t$
convolution (16.3), p. 126	BBD (3.11) and its $s \rightarrow t$ form (proof of Lemma 3.11); BB (1.3)
transition $\mathbb{T}_{c,p}^{h_c}$ (16.4), p. 126	fluctuation measure μ_t^φ , BBD (3.17)–(3.18); kernel of $\mathbf{P}_{s,t}$, BBD (3.12); BB (1.4)
Hessian identity (16.11), p. 129; gradient $\nabla_U \phi_c = \Delta_U^{-1}(\mathbb{E}_\top Y_U - h)$ (§16.4, p. 135)	BBD (3.61), second equality; (3.60)
Brascamp–Lieb bound	BBD (3.63)–(3.64)
$\text{COV}_\top^{\text{ext}}(Y) \preceq \mathbb{E}_\top^{\text{ext}} M_{\text{ext}}(Y)^{-1}$, $M_{\text{ext}} = \Delta_E^{-1} - [K_p^{\text{ext}}]_{EE}$ (§16.2, p. 129)	
margin $\ \Delta_E^{1/2} L_p \Delta_E^{1/2}\ < 1$ (§16.1, p. 127)	$t < t_0$, BBD Ex. 3.15

recursion (16.14)–(16.15), p. 130	BBD (3.64)–(3.65) with $C_t \rightarrow C_0 = p^2 S$, Hess $V_0 \rightarrow -p^{-2} D^\circ$
scalar bound at the end of the proof of Lemma 16.3, p. 139	AKV Thm. 102 with $\gamma = p^2/q$, $T = (1 - x^2)p^2$
iteration over the caps (proof of Theorem 18.6, p. 170)	semigroup property (BBD proof of Lemma 3.11); Hessian flow (3.56); discrete decompositions admitted by BB Thm. 2.6
$\mu_{p,h}$ (more observed) $\rightarrow \mu_{c,h}$, with $K = \text{Cov}$	AKV $\nu_{t+1} \rightarrow \nu_t$, backward induction (8)

The two key steps, side by side:

Preprint	BBD
(16.11) $K_c = \Delta_U^{-1} [\text{Cov}_T(Y)]_{UU} \Delta_U^{-1} - \Delta_U^{-1}$	(3.61) $(f, \text{Hess } V_t f) = (f, C_t^{-1} f) - \text{Var}_{\mu_t^\varphi}((C_t^{-1} f, \zeta))$
(16.11) with the bound on p. 129, $K = [K_p^{\text{ext}}]_{EE}$ $K_c^{\text{ext}} \preceq [\mathbb{E}_{T^{\text{ext}}} (\Delta^{-1} (\Delta^{-1} - K)^{-1} \Delta^{-1} - \Delta^{-1})]_{UU}$ $= [\mathbb{E}_{T^{\text{ext}}} K (I - \Delta K)^{-1}]_{UU}$	(3.64) $\text{Hess } V_t \geq \mathbb{E}_{\mu_t^\varphi} [\text{Hess } V_0 (C_t^{-1} + \text{Hess } V_0)^{-1} C_t^{-1}]$

In the second row put $H = -K$ and $C = \Delta$ on E . BBD’s integrand is then $H(C^{-1} + H)^{-1} C^{-1} = H(I + CH)^{-1} = -K(I - \Delta K)^{-1}$, so (3.64) is the preprint’s bound before compression to U and recentring. The preprint then recentres at the frozen spherical Gaussian, using $M^{-1} = C_0 + p^{-2} C_0 D^\circ (I - SD^\circ)^{-1} C_0$ with $M = C_0^{-1} - p^{-2} D^\circ$ (§16.2, p. 130). The Gaussian part gives $R_c|_U$ exactly, and the correction map $D^\circ \mapsto D^\circ (I - SD^\circ)^{-1}$ in (16.14) is the same map $H \mapsto H(I + CH)^{-1}$ at $H = -p^{-2} D^\circ$, $C = C_0$, up to the factor $-p^{-2}$. In its whitened form (16.15), $D(I - SD)^{-1} = S^{-1/2} [(I - S^{1/2} D S^{1/2})^{-1} - I] S^{-1/2}$, the bracket is $-\hat{H}(\text{id} + \hat{H})^{-1}$, the integrand of BBD (3.65), with $\hat{H} = -S^{1/2} D S^{1/2}$.

What is new in the preprint.

- The iteration of the Brascamp–Lieb step over a discrete sequence of caps. In the sources this bound is a single step (BBD Prop. 3.13 from 0 to t , BBD Ex. 3.15, AKV Thm. 102).
- The radial deficit, regenerated at each step by completing the square, Sherman–Morrison and Gaussian-bridge trace identities ((16.16)–(16.20), pp. 130–131). The sources have no radial or rank-one mechanism.
- Conditional-Haar compression of transverse errors (§16.4, pp. 131–136): sampled replicas, a nuclear-norm integration by parts and a conditionally Haar residual rotation.
- Bounds on typical fields only (annuli and bounded balls), where the sources assume field-uniform bounds. The Brascamp–Lieb margin is restored by the semiconcave extension (16.7), and matching lower bounds come from cutoff Fisher (Cramér–Rao) tests (16.12)–(16.13) and transition comparisons.
- Initialization at a fixed small p_0 from a TAP expansion and an auxiliary potential (§§17–18), not from a product or explicit measure.

- The cap schedule $B_p = (p^2 - d)_+$, i.e. $J_p = \min(J, 2 - p^2)$ spectrally, which flattens the top of the spectrum.
- The endpoint: entropy retention on stopped observation paths, then layers (§19), rather than the multiscale Bakry–Émery criterion.

How to check.

1. Put (16.3)–(16.4), p. 126 beside [9, Def. 3.2, (3.11)–(3.12), p. 16; (3.17), p. 17; proof of Lemma 3.11, p. 22] (or [8, (1.3)–(1.4), p. 2]), and substitute $V_s = -\phi_p$, $C_t - C_s = \Delta$, $\varphi = h_c$.
2. Put (16.11), p. 129 beside [9, Lemma 3.12, (3.61), second equality, p. 23], with $\text{Hess } V = -K$ and $C_t = \Delta$.
3. Combine (16.11) with the Brascamp–Lieb bound on p. 129, inverting on E before compressing to U , and check the identity $\Delta^{-1}(\Delta^{-1} - K)^{-1}\Delta^{-1} - \Delta^{-1} = K(I - \Delta K)^{-1}$, $K = [K_p^{\text{ext}}]_{EE}$, against [9, (3.64), p. 23]. Then put (16.14)–(16.15), p. 130 beside (3.64)–(3.65), with $C_t \rightarrow C_0 = p^2 S$ and $\text{Hess } V_0 \rightarrow -p^{-2} D^\circ$.
4. In the last display of the proof of Lemma 16.3, p. 139, the support curvature on U is q/p^2 with $q = (1 + x^2 a)^{-1} + e$, $a = \alpha/c^2$, and the increment is $(1 - x^2)p^2$. Check that

$$c^2 K_c^{\text{ext}} \preceq \frac{1 + (1 + x^2 a)e}{(1 + a) - (1 - x^2)(1 + x^2 a)e/x^2} I_U = \frac{x^2 q}{1 - (1 - x^2)q} I_U = c^2 (\gamma - T)^{-1} I_U$$

with $\gamma = p^2/q$ and $T = (1 - x^2)p^2$, which is [10, Thm. 102, p. 76].

3.5 S5. The capacity–measure criterion for the log-Sobolev inequality

Summary. Route 2 ends with Lemma 19.1 (p. 172), which turns an energy estimate for a restricted class of bounded tests into the log-Sobolev inequality. After a new first step that yields a lower bound on capacity, its proof (pp. 172–173) runs the sufficiency half of the capacity–measure criterion for the log-Sobolev inequality, stated by Barthe–Roberto [11, §4, remark 6]) and proved by Barthe–Cattiaux–Roberto [12, Thms 11 and 13], with the dyadic slices of Bakry–Coulhon–Ledoux–Saloff-Coste [13, §10.1]. The same slicing, without capacities, recurs in the proof of Theorem 14.2 (pp. 111–112). The preprint cites only Barthe–Roberto’s Lemma 4, as a “compare” pointer for the final entropy shift (p. 173); the capacity, the criterion, the slicing and the median split carry no citation, and the proof of Theorem 14.2 cites nothing. Grade: STRONG, because the whole chain from capacity to the log-Sobolev inequality is reused intact and only its last link is credited.

Setup. For a probability measure η , $\text{Ent}_\eta(u) = \eta[u \log u] - \eta[u] \log \eta[u]$ (§19, p. 171). With KL for the preprint’s $D(\cdot \parallel \cdot)$:

Lemma 19.1 (p. 172). Let η be a strictly positive probability measure on a finite state space, $\mathcal{E}$ a reversible Markov Dirichlet form, and $\min_x \eta(x) \geq e^{-M}$ with $M \geq \log 2$. Suppose that, for some $A > 0$, $\mathcal{E}(g) \geq A \eta[g^2] \log(1/\eta[g^2])$ for every $0 \leq g \leq 1$ supported on a set of measure at most $1/2$ with $\text{KL}(g^2 \eta / \eta[g^2] \parallel \eta) \geq \frac{1}{2} \log(1/\eta[g^2])$ and $0 < \log(1/\eta[g^2]) \leq M$. Then, with an absolute constant C ,

$$\text{Ent}_\eta(f^2) \leq \frac{C(1 + \log_+ M)^2}{A} \mathcal{E}(f) \quad \text{for every real-valued } f. \quad (19.3)$$

For a set E with $\eta(E) = a \leq 1/2$, a test h is *admissible* if $h = 1$ on E and $h = 0$ outside a set of measure at most $1/2$. The preprint defines capacity as the infimum of $\mathcal{E}(h)$ over admissible h

(p. 172); we write $\text{cap}_\eta(E)$ for it. Step 1 of the proof shows

$$\mathcal{E}(h) \geq \frac{cA}{(1 + \log_+ M)^2} a \log(1/a) \quad \text{for every admissible } h. \quad (19.4)$$

In the application $\eta = \mu_J$, $\mathcal{E} = \mathcal{D}_J$, $A = n^{-2/3-r}$, and the hypothesis is the restricted estimate (19.1) for tests satisfying (19.2); on $\|J\| \leq 3$ one has $M = Cn$, and the factor $(\log n)^2$ is absorbed into the exponent (p. 173).

Role in the argument. Lemma 19.1 is the last link of route 2 (Section 2). §§19.2–19.6 prove (19.1) for tests satisfying (19.2) by following the observation process to a product endpoint, which is designed for bounded tests whose squared-density tilt has entropy comparable to $\log(1/\mu_J[f^2])$ (p. 172). Lemma 19.1 turns this into Theorem 19.3, $\text{Ent}_{\mu_J}(f^2) \leq n^{2/3+\rho} \mathcal{D}_J(f)$, which gives the bound on $C_{\text{LS},J}$ in Theorem 1.1 (§20.3, p. 186). The proof has three steps: (1) equal-height layers give (19.4) (p. 172); (2) dyadic layers give (19.3) for $v \geq 0$ supported on a set of measure at most $1/2$ (pp. 172–173); (3) a median split and an entropy shift give (19.3) for all f (p. 173). Steps 2 and 3 use the hypothesis only through (19.4), and they are the reused part. The same slicing, without capacities, removes the sup-norm restriction from the support profile of route 1 in the proof of Theorem 14.2.

The earlier technique. Maz'ya [34] reduced L^2 Sobolev-type inequalities to inequalities between the capacity and the measure of sets, by cutting a function into level slices. On a probability space this takes the following form. (BR, BCR and BCLS abbreviate [11], [12] and [13]; BCR page numbers refer to arXiv:math/0407219v1. In the sources A is a set; ρ and φ are unrelated to the preprint's ρ and ϕ_j , and p, q are BCLS's exponents, unrelated to the preprint's p and p_j .)

- *Capacity of a set of measure at most $1/2$* [12, §5.2, pp. 30–31]: $\text{Cap}_\mu(A) = \inf\{ \int |\nabla f|^2 d\mu : f|_A \geq 1, \mu(f = 0) \geq \frac{1}{2} \}$. Barthe–Roberto [11, §4, remark 6), p. 496] use tests $\mathbf{1}_A \leq f \leq \mathbf{1}_\Omega$ with $\mu(\Omega) = 1/2$.
- *The criterion* [11, remark 6), pp. 495–496], stated without proof: the log-Sobolev constant is comparable, up to universal constants, to the best β in $\beta \text{Cap}_\mu(A) \geq \mu(A) \log(1/\mu(A))$ for $\mu(A) \leq 1/2$. The programme is to split f at a median m into $F_\pm = (f - m)\mathbf{1}_{\{f \gtrless m\}}$ and treat each half by Maz'ya's level-set argument.
- *Its proof* [12, Thm 11, pp. 38–39; Thm 13, p. 45; Rem. 19, pp. 46–47]. Theorem 11: for $\rho > 1$ and a non-decreasing $F \geq 0$ with $F = 0$ on $[0, 2\rho]$, if $\mu(A)F(\rho/\mu(A)) \leq D \text{Cap}_\mu(A)$ whenever $\mu(A) \leq 1/(2\rho)$, then

$$\int f^2 F\left(\frac{f^2}{\int f^2 d\mu}\right) d\mu \leq D \left(\frac{\rho}{\sqrt{\rho}-1}\right)^2 \int |\nabla f|^2 d\mu.$$

Its proof sets $\Omega_k = \{f^2 \geq 2\rho^k \mu(f^2)\}$, bounds the left side by $\sum_{k \geq 1} \mu(\Omega_k) 2\rho^{k+1} \mu(f^2) F(2\rho^{k+1})$, and tests $\text{Cap}_\mu(\Omega_k)$ with h_k , the function $|f|$ clipped between consecutive levels and rescaled to $[0, 1]$. Theorem 13 with $\varphi = \log$, plus Remark 19 (which derives the needed Poincaré inequality from capacity), gives: $\mu(A) \log(2/\mu(A)) \lesssim \text{Cap}_\mu(A)$ implies the log-Sobolev inequality.

- *Dyadic slices* [13, §10.1, p. 1065]: the capacity inequality is applied to $2^{-k} f_k = 2^{-k}[(f - 2^k)^+ \wedge 2^k]$, which is at least 1 on $\{f \geq 2^{k+1}\}$, giving $2^{pk} \mu(f \geq 2^{k+1})^{p/q} \leq C W(f)^p$. Summing needs the slice energies to add up to at most a constant times the energy (hypothesis (H_ρ^p) , p. 1038; (H) for short), which holds for Markov kernel forms (Lemma 7.1, pp. 1055–1057). Equation (10.4) (p. 1069), stated after Theorem 10.4, is the level-set bound for Orlicz norms; Theorem 10.4 is

presented as a generalization of Kaimanovich's Theorem 3.1 for Markov operators [35]. Theorem 10.5 (p. 1070) uses the same slices to upgrade a support-restricted inequality to an Orlicz inequality.

- *Median split and Rothaus shift* [11, Lemma 4, p. 485; proof of Thm 3, pp. 486–487]: $\text{Ent}_\mu(F^2) \leq \text{Ent}_\mu(F_+^2) + \text{Ent}_\mu(F_-^2)$ for the halves at a median, and $\text{Ent}_P(f^2) \leq \text{Ent}_P((f-a)^2) + 2 \int (f-a)^2 dP$ for every $a \in \mathbb{R}$.

Correspondence. The first table below matches the links; the second sets the two key displays side by side.

Preprint	Earlier work
Admissible h : 1 on E , 0 off a set of measure $\leq 1/2$, clipped to $[0, 1]$; $\text{cap}_\eta(E) = \inf \mathcal{E}(h)$ (p. 172)	$\text{Cap}_\mu(A)$: $f _A \geq 1$, $\mu(f=0) \geq 1/2$ (BCR §5.2, p. 31); $\mathbf{1}_A \leq f \leq \mathbf{1}_\Omega$, $\mu(\Omega) = 1/2$ (BR remark 6), p. 496)
(19.4) implies (19.3), constant $\asymp (1 + \log_+ M)^2/A$ (p. 172)	$\beta \text{Cap}_\mu(A) \geq \mu(A) \log(1/\mu(A)) \Leftrightarrow \text{LSI}$ (BR remark 6), p. 496); BCR Thm 13 at $\varphi = \log$ with Rem. 19
Slices $\phi_j = (v - 2^j)_+ \wedge 2^j$, divided by 2^j , test $\{v \geq 2^{j+1}\}$; $a_j = \eta(v \geq 2^j)$ (p. 172)	$2^{-k}[(f - 2^k)_+ \wedge 2^k] \geq 1$ on $\{f \geq 2^{k+1}\}$ (BCLS §10.1, p. 1065); BCR's h_k (p. 39)
$\mathcal{E}(v) \geq \sum_j \mathcal{E}(\phi_j)$ by edgewise monotone increments (pp. 172–173)	(H) for kernel forms (BCLS Lemma 7.1, pp. 1055–1057)
$\text{Ent}_\eta(v^2) \leq 48S$, $S = \sum_j 4^j a_{j+1} \log(1/a_{j+1})$ (p. 173)	Level-set bound in the proof of BCR Thm 11 (p. 39); BCLS (10.4) (p. 1069)
Median b , $v_\pm = (f - b)_\pm$; $\text{Ent}_\eta((f - b)^2) \leq \text{Ent}_\eta(v_+^2) + \text{Ent}_\eta(v_-^2)$ (p. 173)	$F_\pm$ at a median (BR remark 6), p. 495); the same subadditivity (BR proof of Thm 3, p. 487)
(19.5): $\text{Ent}_\eta((u + b)^2) \leq C \text{Ent}_\eta(u^2) + C\eta[u^2]$, cited (p. 173)	BR Lemma 4 (p. 485), which is (19.5) with $C = 2$
§14: $g_j = \min\{(f - 2^j)_+, 2^j\}$; (14.17)–(14.18); Prop. 14.1 applied to each $g_j/\sqrt{A_j}$ and summed (pp. 111–112)	BCLS Thm 10.5 (p. 1070): support-restricted to Orlicz, by a weak bound on each $f_k = (f - 2^k)_+ \wedge 2^k$ summed with (10.4) and (H)

Key displays, proof of Lemma 19.1	Earlier work
Criterion, (19.4) $\Rightarrow$ (19.3) (p. 172): $\text{cap}_\eta(E) \geq \frac{cA}{(1+\log_+ M)^2} a \log \frac{1}{a}$ for $a \leq \frac{1}{2}$ gives $\text{Ent}_\eta(f^2) \leq \frac{C(1+\log_+ M)^2}{A} \mathcal{E}(f)$	BR remark 6), p. 496: $\beta \text{Cap}_\mu(A) \geq \mu(A) \log \frac{1}{\mu(A)}$ for $\mu(A) \leq \frac{1}{2} \Leftrightarrow \text{LSI}$ with constant $\asymp \beta$
Slices and level sets (pp. 172–173): $\mathcal{E}(v) \geq \sum_j \mathcal{E}(\phi_j) \geq \frac{cA}{(1+\log_+ M)^2} S$ and $\text{Ent}_\eta(v^2) \leq 48S$	BCLS p. 1065: $2^{pk} \mu(f \geq 2^{k+1})^{p/q} \leq C W(f)^p$; BCR p. 39: $\int f^2 F(\cdot) d\mu \leq \sum_{k \geq 1} \mu(\Omega_k) 2\rho^{k+1} \mu(f^2) F(2\rho^{k+1})$

What is new in the preprint.

- Step 1: the passage from an entropy-conditioned estimate on bounded tests to the capacity bound (19.4), through equal-height layers $g_i = K((h - i/K)_+ \wedge K^{-1})$ with $K \asymp \log \log(1/a)$ and the doubly exponential pigeonhole $T_{i+1} \geq S_i^2$, at a cost of $(1 + \log_+ M)^2$.
- The criterion stated for a finite reversible Markov Dirichlet form, with the slice superadditivity proved edgewise with constant 1.
- In §14, the likelihood restriction $\|f\|_\infty^2 \leq 100p^{-2}$ of Proposition 14.1 (p. 105) and the rule discarding slices with $A_j < 4^j p_j^2/100$, at a cost of at most $S/50$ (p. 112).

How to check.

1. *Capacity and criterion.* Put the opening of the proof of Lemma 19.1, (19.4) and the sentence after it (p. 172) beside [12, §5.2, p. 31; Thm 13, p. 45; Rem. 19, pp. 46–47], with Theorem 13 at $\varphi = \log$, and [11, remark 6], p. 496].
2. *Slices and level sets.* Compare the dyadic-layer paragraph (pp. 172–173) with [13, §10.1, p. 1065] and the first two displays of the proof of [12, Thm 11, p. 39].
3. *Median split and shift.* Compare p. 173 with [11, proof of Thm 3, p. 487; Lemma 4, p. 485].
4. *§14.* Compare the proof of Theorem 14.2, including (14.17)–(14.18) (pp. 111–112), with [13, Thm 10.5, p. 1070; Lemma 7.1, pp. 1055–1057].

4 Partial findings

4.1 P1. Wang’s signed two-spin curvature method

Summary. The curvature inequality behind the terminal spectral gap of route 1 is Wang’s signed two-spin method [14, Lemma 3.4 and proof of Thm 1.2, pp. 10–13]: condition on all but two spins i, j , expand the two-spin Ising law in $U_{ij} = \tanh B_{ij}$ keeping the signed first-order term, and sum over pairs, so that $\|\mathcal{L}f\|^2$ is at least the Dirichlet form minus $a^\top U a$, with B the zero-diagonal interaction matrix and $a_i = v_i d_i f$, up to higher-order terms in U . The preprint imports it as Lemma 8.1 of AcceptedGap (§3, p. 8) and applies it in the proof of Proposition 13.6 (pp. 102–103). At both places the preprint cites only the companion lemma, whose section opens by crediting Wang’s Lemma 3.4 (AcceptedGap, §8, p. 58); the preprint’s one citation of Wang is for his $\beta < 1/2$ mixing theorem (§1, p. 5). Grade: PARTIAL, because Wang’s two-spin estimate and its summation over pairs are reused essentially intact, but their credit reaches the reader only through the cited companion lemma.

Setup. AG denotes [25]. Let $\mu_h(x) \propto \exp(\frac{1}{2}x^\top Bx + h^\top x)$ on $\{-1, 1\}^n$, with B symmetric with zero diagonal, and let P_i be the heat-bath update at site i (the Q_i of Section 2), an orthogonal projection in $L^2(\mu_h)$. Put

$$D_i = I - P_i, \quad H = \sum_i D_i = -\mathcal{L}, \quad d_i f(x) = \frac{1}{2}(f(x^{i \leftarrow +1}) - f(x^{i \leftarrow -1})),$$

let m_i, v_i be the conditional mean and variance of x_i given x_{-i} , and $a_i = v_i d_i f$. Then $D_i f = (x_i - m_i) d_i f$ and $\mathcal{D}_h(f) = \sum_i \mu_h[v_i(d_i f)^2] = \sum_i \|D_i f\|_{L^2(\mu_h)}^2$. Entrywise, $U = \tanh B$ (Wang’s T), $Q = U \circ U$, $r_3(U) = \max_j \sum_j |U_{ij}|^3$ and $r_4(U) = \max_i \sum_j U_{ij}^4$. In Proposition 13.6, $B = \theta J^\circ$ with $0 \leq \theta \leq 1$, and h is an observation field at a large fixed time T or later.

Role in the argument. Route 1 needs, at a large fixed observation time, a posterior gap that holds uniformly over the law of the observed configuration X , and bounded covariance and gap along a path that weakens the interaction to a product law. Proposition 13.6 supplies both, outside probability e^{-cn} . Its proof restates the imported inequality (p. 102), checks its entrywise hypothesis $\max |U_{ij}| \leq u_0$ for every $B = \theta J^\circ$, bounds $\mu_h[a^\top Ua + C|a|^\top Q|a|]$ by $\mathcal{D}_h(f)/4$ through a large-field split (Section 4.6), and concludes (p. 103) from $\mu_h[(-\mathcal{L}f)^2] \geq \mathcal{D}_h(f)/2$, by the spectral theorem, that every nonzero eigenvalue of $-\mathcal{L}$ is at least $1/2$, so $\text{Var}_{\mu_h} f \leq 2\mathcal{D}_h(f)$ and $\|\Sigma_h\| \leq 2$. These outputs feed the backward amplification (§12.3, p. 71) and the proof of Proposition 14.1 (pp. 109, 111).

The earlier technique. Wang [14] (arXiv v1, 23 August 2026; pages are the printed pages of v2) proves a deterministic Poincaré criterion (Thm 1.2, p. 3) for Glauber dynamics of Ising models $\mu_{J,h}$ with weak entries, uniformly in h : with $T = (\tanh J_{ij})$, the constant is $\frac{1}{n}(1 - \max\{\|T\|, 2\|T \circ T\|\} - Cr_3(T))$ whenever this is positive. His generator is $\frac{1}{n} \sum_i (P_i - I) = -H/n$, with $D_i = I - P_i$.

- *Curvature form* (Prop. 3.2, p. 9; Lemma 3.3, p. 10): $n^2 \mathbb{E}_\mu \Gamma_2(f, f) = \|\sum_i D_i f\|^2 = \sum_{i,j} \langle D_i f, D_j f \rangle$, $D_i f = (x_i - m_i) d_i f$ and $\|D_i f\|^2 = \mathbb{E}_\mu [v_i (d_i f)^2]$.
- *Two-spin estimate* (Lemma 3.4, p. 10; proof in §3.2, pp. 11–13), which Wang presents as his new ingredient (§1.1, p. 2). If $\mu(x) \propto \exp(J_{12}x_1x_2 + h_1x_1 + h_2x_2)$ and $t = \tanh J_{12}$ has $|t| \leq \delta$, then for every f

$$\langle D_1 f, D_2 f \rangle \geq -t \langle v_1 d_1 f, v_2 d_2 f \rangle - 2t^2 \langle m_1 v_1 d_1 f, m_2 v_2 d_2 f \rangle - C|t|^3 (\|D_1 f\|^2 + \|D_2 f\|^2).$$

The proof, with $a = \tanh h_1$, $b = \tanh h_2$, $A = 1 - a^2$, $B = 1 - b^2$ and $f = c + p(x_1 - a) + q(x_2 - b) + r(x_1 - a)(x_2 - b)$, writes the left side plus the two signed terms as $AB(\phi_0 pq + \phi_1 pr + \phi_2 qr + \phi_{12} r^2)$ with $|\phi_0| \leq C_0|t|^3$, $|\phi_1| + |\phi_2| \leq C_0 t^4$, $\phi_{12} \geq \frac{1}{2}$; its steps are in the table below.

- *Summation* (proof of Thm 1.2, pp. 10–11): condition on $X_{-\{i,j\}}$, apply Lemma 3.4 to the conditional pair law, average, and sum over pairs. With $a = (v_i d_i f)_i$ and $b = (m_i v_i d_i f)_i$,

$$n^2 \mathbb{E}_\mu \Gamma_2(f, f) \geq (1 - 2Cr_3(T)) \sum_i \|D_i f\|^2 - \mathbb{E}_\mu [a^\top T a + 2b^\top (T \circ T) b].$$

Wang bounds the quadratic forms by $\max\{\|T\|, 2\|T \circ T\|\} \sum_i \|D_i f\|^2$ and converts curvature to a gap by the integrated Bakry–Émery criterion (Thm 2.11, p. 8), which he quotes from [36, Prop. 4.8.3]. For SK, $\|T\| \approx 2\beta$, which gives a gap for $\beta < 1/2$ (Thm 3.1, p. 8).

Correspondence. Wang’s steps, against AG’s Lemma 8.1 and its proof as imported by the preprint. In AG, $g_i = d_i f$, $a_i = v_i^1 g_i$, and $u = \tanh J_{ij}$ plays the role of t .

Step	Wang	AG (and the preprint)
Curvature form	$n^2 \mathbb{E}_\mu \Gamma_2 = \ \sum_i D_i f\ ^2$; $\sum_i \ D_i f\ ^2 = \sum_i \mathbb{E}_\mu [v_i (d_i f)^2]$ (Prop. 3.2, Lemma 3.3, pp. 9–10)	$\mathbb{E}_{\mu_{h_0}} [(Hf)^2]$, $Hf = \sum_i (I - P_i) f = \sum_i (x_i - m_i^1) d_i f$; $\sum_i \mathbb{E} [v_i^1 g_i^2] = \mathcal{D}_{h_0}(f)$ ((8.1), p. 59; p. 61). Preprint: $\mu_h[(-\mathcal{L}f)^2]$, $\mathcal{D}_h(f)$ (p. 102)

Pair reduction	Condition on $X_{-\{i,j\}}$; $\mu \propto (1 + ax_1)(1 + bx_2)(1 + tx_1x_2)$; $v_1 = A(1 - t^2)/(1 + atx_2)^2$; $d_1f = p + r(x_2 - b)$, $d_2f = q + r(x_1 - a)$ (pp. 11–12)	Condition on the other spins; $d\mu_{ij}/d\mu_{0,ij} = (1 + uxy)/(1 + uab)$; $v_i^1 = A(1 - u^2)/(1 + uay)^2$; $g_i = p + r(y - b)$, $g_j = s + r(x - a)$ ((8.3)–(8.4), p. 59)
Signed first-order term	Add $t\langle v_1d_1f, v_2d_2f \rangle$ and $2t^2\langle m_1v_1d_1f, m_2v_2d_2f \rangle$: pq coefficient $O(t^3)$, pr, qr coefficients $O(t^4)$, r^2 coefficient $\geq \frac{1}{2}$ (p. 12)	Add $u\mathbb{E}_{ij}[a_ia_j]$ (8.8) to the cross term (8.6): ps, rp, rs coefficients $O(u^2)$, r^2 coefficient $\geq \frac{1}{2}$ (pp. 60–61)
Error terms	$\frac{1}{2}r^2$ dropped; the left side plus the two signed terms is $\geq -C_0AB t ^3(p^2 + q^2 + r^2)$; $\ D_1f\ ^2 \geq C_1(Ap^2 + AB r^2)$ (pp. 12–13)	Young into $\frac{1}{2}r^2$, leaving $C_5u^2 ps + C_5u^4(p^2 + s^2)$; $\mathbb{E}_{ij} a_ia_j \geq c_1AB ps $, $\mathbb{E}_{ij}[v_i^1g_i^2] \geq c_2ABp^2$ (p. 61)
Sum over pairs	For $\max T_{ij} \leq \delta$: $a^\top Ta + 2b^\top (T \circ T)b$; error $2Cr_3(T) \sum_i \ D_if\ ^2$ (p. 11)	For $\max U_{ij} \leq u_0$: $a^\top Ua + C a ^\top Q a $; error $Cr_4(U)\mathcal{D}_{h_0}(f)$ ((8.2), pp. 59, 61); restated in the preprint (p. 102)

The two key formulas side by side; the second AG formula is the inequality the preprint states on p. 102, with $\mu_h[(-\mathcal{L}f)^2]$ on the left:

Wang	AG (and the preprint)
Lemma 3.4 (p. 10): $\langle D_1f, D_2f \rangle \geq -t\langle v_1d_1f, v_2d_2f \rangle -$ $2t^2\langle m_1v_1d_1f, m_2v_2d_2f \rangle - C t ^3(\ D_1f\ ^2 + \ D_2f\ ^2)$	(8.9), p. 61: $\mathbb{E}_{ij}[(x - m_i^1)g_i(y - m_j^1)g_j] \geq$ $-u\mathbb{E}_{ij}[a_ia_j] - C_6u^2\mathbb{E}_{ij}[a_ia_j] - C_6u^4\mathbb{E}_{ij}[v_i^1g_i^2 + v_j^1g_j^2]$
Proof of Thm 1.2 (p. 11): $n^2\mathbb{E}_\mu\Gamma_2(f, f) \geq$ $(1 - 2Cr_3(T)) \sum_i \ D_if\ ^2 - \mathbb{E}_\mu[a^\top Ta + 2b^\top (T \circ T)b]$	(8.2), p. 59: $\mathbb{E}_{\mu_{h_0}}[(Hf)^2] \geq$ $(1 - Cr_4(U))\mathcal{D}_{h_0}(f) - \mathbb{E}_{\mu_{h_0}}[a^\top Ua + C a ^\top Q a]$

What is new in the preprint.

- The imported variant, proved in AG (pp. 59–61), keeps only the first-order term signed: its second-order cost is the absolute form $C|a|^\top Q|a|$ rather than Wang’s signed $2b^\top (T \circ T)b$ with $b_i = m_ia_i$, and absorbing the mixed terms into $\frac{1}{2}r^2$ by Young’s inequality makes the row error quartic, $r_4(U)$, rather than cubic, $r_3(T)$.
- The quadratic forms are bounded differently. Wang bounds $a^\top Ta$ by $\|T\| \sum_i \|D_if\|^2$, which needs $\|T\| < 1$; at $\beta = 1$, $\|J^\circ\| \approx 2$. The preprint bounds $a^\top Ua + C|a|^\top Q|a|$ pointwise by a large-field split at observation time T (Section 4.6), as AG do at fixed $j < 1$.
- The inequality is applied along the family $B = \theta J^\circ$, $0 \leq \theta \leq 1$, that ends at a product law, with disorder events of probability $1 - e^{-cn}$; AG apply it at one fixed $j < 1$.

How to check.

1. *Pair estimate.* Put [14, Lemma 3.4, p. 10] and its proof (pp. 11–13) beside AG (8.3)–(8.9) (pp. 59–61): the same pair law, the same p, r expansion of the two gradients, the same signed term.

2. *Summation.* Compare the display that follows the summation over pairs in the proof of Wang’s Thm 1.2 (p. 11) with AG (8.2) (p. 59), the last paragraph of AG’s proof (p. 61), and the preprint’s restatement (p. 102).
3. *Citation path.* Read the preprint’s §3 (p. 8) and the first paragraph of the proof of Proposition 13.6 (p. 102); then AG §8 (p. 58) and AG §1.1 (p. 4); then the preprint’s only citation of Wang (§1, p. 5).

4.2 P2. Density bounds for weighted chi-square sums (Bobkov–Naumov–Ulyanov; Götze–Naumov–Spokoiny–Ulyanov)

Summary. The preprint conditions Gaussians on spheres using two published density estimates for weighted noncentral chi-square sums: the Fourier majorant of Bobkov, Naumov and Ulyanov (BNU below) [15, Lemma 3, p. 5], and the bound of Götze, Naumov, Spokoiny and Ulyanov (GNSU below) on the density of $\|\xi - a\|^2$ by $\|\Sigma_\xi\|_{\mathbb{F}}^{-1}$, uniformly in the shift a , when no eigenvalue of Σ_ξ dominates ($3\lambda_1^2 \leq \|\Sigma_\xi\|_{\mathbb{F}}^2$) [16, Thm 2.6, p. 13]. The majorant is the dominating function in the proof of Lemma 5.2 (p. 17) and is derived explicitly, by BNU’s concavity argument, in the proof of Lemma 10.1 (p. 46). The shift-uniform bound is the estimate $f_{S|Z=z}(n) \leq C\sqrt{d/n}$ “uniformly in the translated mean” in the shell conditioning of §5.3 (p. 18), and the noncentral bound drawn from (15.5). Neither paper is cited anywhere in the preprint. The companion [24] credits BNU’s Lemma 3 for the same majorant in its own “Density at the mean” lemma (Lemma 4.2, p. 16), but the preprint proves its lemma itself and cites the companion only for its Theorems 1.2–1.3. Grade: PARTIAL, because only a fragment of the two papers is reused: the upper density bound for spread spectra (BNU’s Lemma 3, and the bound of BNU’s Theorem 2 and case 1 of GNSU’s Theorem 2.6). Their lower bounds and dominant-eigenvalue cases are not used.

Setup. For a fixed spectrum and field, $\lambda > \lambda_{\max}(J)$ solves (5.4)

$$D = (\lambda - J)^{-1}, \quad m_o = Dh, \quad \text{Tr } D + \|m_o\|^2 = n,$$

and the spherical Gibbs law is $N(m_o, D)$ conditioned on $\|x\|^2 = n$ (§5.3, p. 17). When $\lambda - 2 \asymp d^2$ with $d_0 \leq d \lesssim 1$, one has $\text{Tr } D^2 \asymp n/d$ and $\|D\|/(\text{Tr } D^2)^{1/2} = O((nd^3)^{-1/2}) = o(1)$ (pp. 17–18).

- *Lemma 5.2 (Density at the mean)* (p. 17). Let g be a finite-dimensional standard Gaussian vector and $Q = g^\top A g + b^\top g$ with $A \succeq 0$ and $\sigma^2 = \text{Var } Q > 0$. The density of Q at $\mathbb{E}Q$ is at least c/σ for an absolute $c > 0$, and, with a smaller constant, at a displacement $o(\sigma)$ along any sequence of such forms. The proof normalizes to $\sum_i a_i(g_i^2 - 1) + 2\sum_i b_i g_i$, $a_i \geq 0$, $2\sum_i a_i^2 + 4\sum_i b_i^2 = 1$; Fourier inversion handles the case in which neither a single a_i nor the linear part dominates.
- *Shell conditioning* (p. 18). Put $x = m_o + D^{1/2}g$, $S = \|x\|^2$, $Z = V^\top g$ for a matrix V with a fixed number of orthonormal columns, and $P = I - VV^\top$. By (5.4), $\mathbb{E}S = n$. Then $p_{Z|S=n}(z) = p_Z(z) f_{S|Z=z}(n)/f_S(n)$. Given $Z = z$, x is Gaussian with mean $m_o + D^{1/2}Vz$ (the “translated mean”) and covariance $D^{1/2}PD^{1/2}$, whose nonzero eigenvalues are those of the compression $PDP|_{\text{ran } P}$.

Role in the argument. Shell conditioning transfers Gaussian moment and tail bounds for fixed projections to the sphere through the density ratio. Lemma 5.2 bounds the denominator below, $f_S(n) \geq c\sqrt{d/n}$, and the shift-uniform estimate bounds the numerator above, $f_{S|Z=z}(n) \leq C\sqrt{d/n}$ for every z . The bounded ratio, with a local limit dominated by the same majorant (p. 18), gives (5.6), $\|\mathbb{E}_{\text{shell}}[D^{-1/2}(x - m_o)(x - m_o)^\top D^{-1/2}]\| \leq 1 + o(1)$, which §11 uses for its shell density

denominators (pp. 56–57). The two estimates recur in the proof of Proposition 7.1 (p. 27) and in §7.1 (p. 29), in the proof of Lemma 10.1 (p. 46), in the proofs of Proposition 11.1 (p. 56) and Lemma 11.4 (p. 64), in §15.3 ((15.5)–(15.6), p. 115) and the proof of Proposition 15.1 (p. 122), and in the proof of Lemma 18.1 (p. 161).

The earlier technique. BNU write $M(\cdot)$ for the maximum of a density, $W_a = \sum_k \lambda_k (Z_k - a_k)^2$ with $\lambda_1 \geq \dots \geq \lambda_n > 0$ and Z_k independent standard normal, $A_1 = \sum_k \lambda_k^2$ and $B_1 = \sum_k \lambda_k^2 a_k^2$ [15, §2, p. 3].

- *Lemma 3* (p. 5). If $\alpha_1^2 + \dots + \alpha_n^2 = 1$ and $\alpha_k^2 \leq 1/m$ for some $m \in \{1, 2, \dots\}$, the characteristic function of $W = \sum_k \alpha_k Z_k^2$ satisfies, in their eq. (6),

$$|f(t)| \leq (1 + 4t^2/m)^{-m/4}.$$

Proof (pp. 5–6): $-\log |f(t)| = \frac{1}{4} \sum_k \log(1 + 4\alpha_k^2 t^2)$ is concave in $(\alpha_k^2)_k$ and is minimized at the extreme points of $\{0 \leq b_k \leq 1/m, \sum_k b_k = 1\}$. Inversion then bounds the density, $M(W) \leq \frac{1}{2\pi} \int |f|$ (p. 6).

- *Theorem 2* (p. 3) and its proof (pp. 9–11). A shift only lowers the modulus: $|\mathbb{E} e^{it(Z-a)^2}| = (1+4t^2)^{-1/4} \exp\{-2a^2 t^2/(1+4t^2)\}$ (p. 10). If $\lambda_1^2 \leq A_1/3$, then $M(W_a) \leq 2/\sqrt{A_1 + B_1} \leq 2/\sqrt{A_1}$.

GNSU write $p_\xi(x, a)$ for the density of $\|\xi - a\|^2$, where ξ is a centered Gaussian element with covariance Σ_ξ , eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$ and $\Lambda_1^2 = \sum_j \lambda_j^2$.

- *Theorem 2.6* (p. 13), eq. (2.3). For any a , $\sup_{x \geq 0} p_\xi(x, a) \lesssim \kappa(\Sigma_\xi)$, where $\kappa(\Sigma_\xi) = \Lambda_1^{-1} = \|\Sigma_\xi\|_F^{-1}$ when $3\lambda_1^2 \leq \Lambda_1^2$ (eq. (2.1), p. 10). In this case the proof (p. 16) bounds each factor by $(1 + 4\lambda_j^2 t^2)^{-1/4}$ whatever the shift (eq. (3.2)), inverts, and integrates the product with Lemma A.2 (pp. 21–22).

Correspondence.

Step	Preprint	BNU / GNSU
Variable	$\sum_i a_i (g_i^2 - 1) + 2 \sum_i b_i g_i$ (p. 17); S given $Z = z$, the squared norm of a Gaussian with covariance $D^{1/2} P D^{1/2}$ and translated mean (p. 18)	$W = \sum_k \alpha_k Z_k^2$ (Lemma 3); for the shell, W_a (Thm 2) and $\ \xi - a\ ^2$ (GNSU Thm 2.6)
Fourier majorant	$(1 + 4(\max_i a_i)^2 t^2)^{-c/(\max_i a_i)^2}$ (p. 17); concavity of $\log(1 + ux)$ on $[0, \max_i a_i^2]$ (p. 46)	Eq. (6) with $1/m = \max_k \alpha_k^2$; concavity on $\{0 \leq b_k \leq 1/m\}$ (Lemma 3, pp. 5–6)
Shift or linear part dropped	“The noncentral factor does not increase the modulus” (p. 27); (15.5), “the remaining factors have absolute value at most one” (p. 115)	BNU p. 10; GNSU eq. (3.2), p. 16
Spread hypothesis	$\ D\ /(\text{Tr } D^2)^{1/2} = o(1)$ (p. 18); a band of order k_p coordinates with $r_i \asymp p^{-2}$ (p. 115)	$\lambda_1^2 \leq A_1/3$ (BNU Thm 2); $3\lambda_1^2 \leq \Lambda_1^2$ (GNSU eq. (2.1))
Shift-uniform density bound	$f_{S Z=z}(n) \leq C\sqrt{d/n}$ “uniformly in the translated mean” (p. 18); parent density $\leq C\sqrt{p/n}$ “for every y ” (p. 122); $\leq Cp^2/\sqrt{k_p}$ for a tilted law with any mean (p. 161)	$\sup_x p_\xi(x, a) \lesssim \Lambda_1^{-1}$ for any a (GNSU Thm 2.6); $M(W_a) \leq 2/\sqrt{A_1}$ (BNU Thm 2)

The two key formulas, side by side:

- *Majorant.* BNU's eq. (6), applied to $\alpha_k = a_k/s$ with $s^2 = \sum_j a_j^2$ at frequency st , with $m = s^2/\max_k a_k^2$, reads $|\mathbb{E} e^{it \sum_k a_k g_k^2}| \leq (1 + 4(\max_k a_k)^2 t^2)^{-s^2/(4 \max_k a_k^2)}$; a linear term only lowers the modulus. BNU state eq. (6) for integer m ; the chord bound $\log(1+4bt^2) \geq mb \log(1+4t^2/m)$ on $[0, 1/m]$ gives it for real m . Preprint, Lemma 5.2 (p. 17): $(1 + 4(\max_i a_i)^2 t^2)^{-c/(\max_i a_i)^2}$, i.e. $c = s^2/4$. Preprint, Lemma 10.1 (p. 46), with $\sigma^2 = 2 \sum_i a_i^2$: $\prod_i (1 + 4t^2 a_i^2/\sigma^2)^{-1/4} \leq (1 + 4t^2 a_{\max}^2/\sigma^2)^{-\sigma^2/(8a_{\max}^2)}$, which is the display above at frequency t/σ .
- *Shift-uniform bound.* GNSU eq. (2.3) with $\Sigma_\xi = D^{1/2} P D^{1/2}$ and $a = -(m_o + D^{1/2} V z)$: since the compression keeps the spread, $\Lambda_1^2 = \text{Tr}[(PDP)^2] \asymp \text{Tr} D^2 \asymp n/d$, so $\sup_x p_\xi(x, a) \lesssim \sqrt{d/n}$ for every z . Preprint (p. 18): “the surviving spectral block of the compression and the quadratic-product formula therefore give $f_{S|Z=z}(n) \leq C\sqrt{d/n}$ uniformly in the translated mean”.

What is new in the preprint.

- The lower bound at the mean, rather than at the maximum, for every $A \succeq 0$, including a dominant coefficient, a linear term and $o(\sigma)$ displacements. BNU's lower bounds (Lemma 2, p. 4; Theorem 1, p. 3) concern the maximum $M(W)$. The companion's Lemma 4.2 (p. 16) is the centered case.
- The density-ratio transfer of Gaussian projections to the shell, and the sharp constant in (5.6) from a local limit dominated by the majorant.
- Multi-replica and matrix-frequency forms: the 2×2 bound $|\varphi(T)| \leq (1 + c\|T\|^2 \log^{-C} n)^{-cN}$ in the proof of Proposition 7.1 (p. 27), (15.6) (p. 115) and the determinant damping $(1 + c|v|^2/N)^{-c'N}$ in the proof of Lemma 11.4 (p. 64).

How to check.

1. Read the last paragraph of the proof of Lemma 5.2 (p. 17) against BNU Lemma 3 and eq. (6) (p. 5), then the companion's sentence before its Lemma 4.2 (AcceptedCritical, p. 16), which credits BNU's Lemma 3 for this majorant, and its version of the majorant (p. 17).
2. Read “To check the Fourier bound explicitly” in the proof of Lemma 10.1 (p. 46) against the proof of BNU Lemma 3 (pp. 5–6): the same concavity step and the same exponent.
3. Read the shell-conditioning paragraph (p. 18) against GNSU Theorem 2.6 (p. 13), eq. (2.1) (p. 10) and case 1 of its proof (p. 16), and against BNU Theorem 2 (p. 3).
4. Read (15.5) and the paragraph after it (p. 115) against BNU's modulus formula (p. 10): for $Y_i = a_i + \sqrt{r_i} g_i$, BNU's factor at frequency $r_i t$ with shift $a_i/\sqrt{r_i}$ is $(1 + 4t^2 r_i^2)^{-1/4} \exp\{-2t^2 r_i a_i^2/(1 + 4t^2 r_i^2)\}$, the i th factor of (15.5).

4.3 P3. The fixed-spectrum Haar second-moment comparison (Bhattacharya–Sen; Du–Huang)

Summary. Section 6 of the preprint compares cube and sphere partition functions on a Haar orbit at fixed spectrum by a second-moment argument assembled from two earlier pieces: the reweighted-overlap identity and overlap-cell comparison of Du–Huang (DH below) [6, Lemmas 2.1 and 3.9, proof of Thm. 1.6], and, for large overlaps, the fixed-spectrum two-eigenvalue reduction of Bhattacharya–Sen (BS below) [17, §3.2.1, eq. (3.14)]. In the preprint they appear as display (6.3) and the global pair estimate of §6.1 (p. 22), which give $\mathbb{E}_U e^{2F} \leq e^{o(k)}$ in §6.2 (p. 23); the comparison recurs in §§7, 10, 11, 14, 15 and 18. For this comparison Section 6 cites only Comets, for the first moment (its only other citation is Lehec, for the entropy–drift representation). DH

is cited for this content elsewhere (§1, pp. 5–6; §7, p. 26; §12.4, p. 77; §15.4, p. 116), in §7 and §15.4 just before recurrences of the comparison. BS is cited nowhere, and the companion [24], which credits BS for this step (§5, p. 22), is cited by the preprint only for its Theorems 1.2–1.3. **Grade: Partial**, because the DH components are credited elsewhere in the preprint, though not in Section 6, and the uncredited part is BS’s reduction through (3.14), a fragment of their proof.

Setup. In §6 the eigenvalues Λ of J and the eigen-coordinates c of the field are fixed, and the eigenframe U is Haar distributed: $J = U\Lambda U^\top$, $h = Uc$ (p. 20). $\mathbb{E}_U$ is the Haar expectation, $Z(h)$ and $Z_o(h)$ are as in Section 2, $Z_o(h)$ does not depend on U , and $F(U) = \log(Z(h)/Z_o(h))$. For replicas $X^1, \dots, X^p$ (independent uniform signs, or independent uniform points of the sphere of radius $\sqrt{n}$), $R_{ab} = (X^a)^\top X^b/n$, and $R = (R_{ab})$ is their Gram matrix. Proposition 6.1 assumes a spherical saddle λ with $d = \sqrt{\lambda - 2}$ comparable to a scale $r > 0$, $\|h\| \leq Cr\sqrt{n}$ and $k = nr^3 \gg \log n$; μ_h is the cube Gibbs law with field h .

Role in the argument. Proposition 6.1 (pp. 20–21) states that, outside Haar mass e^{-Bk} , $|\log Z(h)/Z_o(h)| \leq o(k)$ (6.1), that tilted p -replica moments cost at most $C \text{Tr}(P^2)k + o(k)$ (6.2), and that pair overlaps concentrate. Section 6.1 bounds the unnormalized replica integrals relative to $Z_o(h)^p$: small overlaps by (6.3) and the common angular integral, and $|R_{12}| > \eta$ by the global pair estimate, which gives $\mathbb{E}_U[e^{2F}q_\eta] \leq e^{-c_\eta n}$ with $q_\eta = \mu_h^{\otimes 2}(|R_{12}| > \eta)$ (p. 23). Section 6.2 turns $\mathbb{E}_U e^F = 1$ and $\mathbb{E}_U e^{2F} \leq e^{o(k)}$ into high anchors, a strong upper tail of F and, through controlled rotations, the lower tail and (6.1). Proposition 6.1 is used in §11 (p. 47), §12.3 (pp. 73, 76) and §14 (pp. 106–107). The comparison itself recurs in §7 (p. 27), in the fixed-overlap second moment of §10.2 (pp. 44–45), in Lemma 11.2 (p. 60) and the proof of Proposition 11.1 (p. 56), where the Gram eigenvalues are $1 \pm t$, in the proof of Proposition 14.1 (p. 106), in the proof of Proposition 15.1 (pp. 117–118), which redoes the Gram comparison as (15.10) and the sum/difference reduction with spherical free energies at $1 \pm |r|$, and in §18.1 (p. 161).

The earlier technique. BS [17, (1.16)–(1.17), p. 5] fix a diagonal matrix Λ , write $\mathbb{E}_0$ for the Haar expectation over O , and study

$$Z_N(\beta, O, \Lambda) = 2^{-N} \sum_{\sigma \in \{\pm 1\}^N} \exp(\beta \sigma^\top O \Lambda O^\top \sigma), \quad \mathbb{E}_0 Z_N = \mathbb{E}_0 \exp(N\beta(O\Lambda O^\top)_{11}).$$

In the proof of Proposition 3.1 (§3.2.1, p. 10), with $\mathbb{E}_1$ the uniform average over σ, τ ,

$$\mathbb{E}_0 Z_N^2 = \mathbb{E}_1 \mathbb{E}_0 \exp\left(N\beta \left\{ \left(1 + \frac{\sigma^\top \tau}{N}\right)(O\Lambda O^\top)_{11} + \left(1 - \frac{\sigma^\top \tau}{N}\right)(O\Lambda O^\top)_{22} \right\}\right),$$

since “the non-zero eigenvalues of $(\sigma\sigma^\top + \tau\tau^\top)/N$ are $(1 + \sigma^\top \tau/N)$ and $(1 - \sigma^\top \tau/N)$ respectively”. Summing the spins by $\mathbb{E}_1 e^{\lambda \sigma^\top \tau} = (\cosh \lambda)^N$ gives (3.14): $\mathbb{E}_0 Z_N^2 = \mathbb{E}_0 \exp(NF_{\text{BS}}(V_1, V_2))$ with $F_{\text{BS}}(x, y) = \beta(x + y) + \log \cosh \beta(x - y)$ and $V_i = (O\Lambda O^\top)_{ii}$.

DH [6, §1, §2.1] take $H_N(x) = \frac{1}{2}(Wx, x)$ with W a GOE matrix, $Z_N = 2^{-N} \sum_x e^{H_N(x)}$, Z_N^{sph} the same integral over the uniform measure on the sphere of radius $\sqrt{N}$, and $X_N = Z_N/Z_N^{\text{sph}}$; $\mathbb{E}X_N = 1$ because Z_N^{sph} averages Z_N over rotations of W ((2.3), p. 6). Lemma 2.1 (p. 7): $J_{\text{DH}}(q) = \mathbb{E}[e^{H_N(x)+H_N(y)} / (Z_N^{\text{sph}})^2]$ depends on x, y only through $q = R(x, y)$; $J_{\text{DH}}(q) = e^{Nq^2/2} e^{N/2} K(q)$ with $K(q) = \mathbb{E}[Z_N^{\text{sph}}(W + (1+q)e_1e_1^\top + (1-q)e_2e_2^\top)^{-2}]$; and $\mathbb{E}[X_N^2 \langle g(R_{1,2}) \rangle] = \mathbb{E}_q[J_{\text{DH}}(q)g(q)]$ for cube overlaps, $\mathbb{E}\langle g(R_{1,2}) \rangle^{\text{sph}} = \mathbb{E}_q^{\text{sph}}[J_{\text{DH}}(q)g(q)]$ for sphere overlaps ((2.7)–(2.8)). The overlap laws are $p(q) = 2^{-N} \binom{N}{N(1+q)/2}$ and $\rho(q) \propto (1 - q^2)^{(N-3)/2}$ ((2.10)–(2.11), p. 7). Lemma 3.9 (p. 12): for

$|q| = o(N^{-1/4})$ and $|\tilde{q} - q| \leq N^{-1}$, $p(q)e^{Nq^2/2} = \rho(\tilde{q})e^{N\tilde{q}^2/2} \cdot \frac{2}{N} \cdot (1 + O(N^{-1} + q^2 + Nq^4))$. The proof of *Theorem 1.6* (pp. 13–14) writes $\mathbb{E}(X_N - 1)^2 = \mathbb{E}_q[J_{\text{DH}}] - \mathbb{E}_q^{\text{sph}}[J_{\text{DH}}]$, compares central cells by Lemma 3.9, and bounds the tails by log-concavity of K (Lemmas 3.3 and 3.7), giving $\mathbb{E}(X_N - 1)^2 \lesssim N^{-1/3}$ (p. 5).

Correspondence.

Step	Preprint	BS	DH
Fixed spectrum, Haar frame	$J = U\Lambda U^\top$, $h = Uc$, with Λ, c fixed (p. 20)	$O\Lambda O^\top$, Λ fixed, $\mathbb{E}_0$ over Haar O ((1.16), p. 5)	GOE W , averaged; rotation invariance (Lemma 2.1)
First moment	$\mathbb{E}_U e^F = 1$ (p. 23; Comets cited on p. 22)	$\mathbb{E}_0 Z_N = \mathbb{E}_0 \exp(N\beta(O\Lambda O^\top)_{11})$ ((1.17))	$\mathbb{E} X_N = 1$ ((2.3), p. 6)
One angular integral for both priors	“Conditional on the Gram matrix, Haar rotation supplies exactly the same angular integral for both priors” (p. 21)	Inner $\mathbb{E}_0$ depends on σ, τ only through $\sigma^\top \tau / N$ (p. 10)	$J_{\text{DH}}(q)$ depends only on q ; (2.7)–(2.8) (p. 7)
Small overlaps: cell comparison	(6.3); spherical density $\propto (\det R)^{(n-p-1)/2}$; Legendre match through cubic order (p. 21)	None	Lemma 3.9 (p. 12); $\rho \propto (1 - q^2)^{(N-3)/2}$ ((2.11))
Large overlaps: two directions	Orthogonal sum and difference rotated; strengths $1 \pm R_{12}$ (p. 22)	Eigenvalues $1 \pm \sigma^\top \tau / N$; V_1, V_2 (p. 10)	$(1 \pm q)e_{1,2}e_{1,2}^\top$ in $K(q)$ ((2.6)); tails by log-concavity (Lemmas 3.3, 3.7)
Spin-overlap cost	Cramér rate of the sign overlap exceeds $R_{12}^2/2$ by a fixed amount on $ R_{12} \geq \eta$ (p. 22)	$\mathbb{E}_1 e^{\lambda \sigma^\top \tau} = (\cosh \lambda)^N$, the log cosh term of F_{BS} in (3.14)	$p(q)$ in (2.10)
Output	$\mathbb{E}_U e^{2F} \leq e^{o(k)}$ (p. 23)	Prop. 3.1: second-moment rate twice the first (p. 9)	Thm. 1.6: $\mathbb{E}(X_N - 1)^2 \lesssim N^{-1/3}$ (p. 5)

The two key formulas, side by side:

- *Two-eigenvalue reduction.* BS (p. 10): the exponent is $N\beta\{(1 + \sigma^\top \tau / N)V_1 + (1 - \sigma^\top \tau / N)V_2\}$. Preprint (p. 22): “After rotating the orthogonal sum and difference, their angular energy is at most” $\exp\{\frac{n}{4}[(1 + R_{12})^2 + (1 - R_{12})^2] + o(n)\}$. Both rest on $xx^\top + yy^\top = (1 + R_{12})w_+w_+^\top + (1 - R_{12})w_-w_-^\top$, where $R_{12} = x^\top y / n$ and $w_\pm = (x \pm y) / \sqrt{2(1 \pm R_{12})}$ are orthogonal with norm $\sqrt{n}$.
- *Cell comparison.* DH (p. 12): $p(q)e^{Nq^2/2} = \rho(\tilde{q})e^{N\tilde{q}^2/2} \cdot \frac{2}{N} \cdot (1 + O(N^{-1} + q^2 + Nq^4))$ with $\rho \propto (1 - q^2)^{(N-3)/2}$. Preprint (6.3), p. 21: cube probability of a Gram box $\leq$ spherical probability $\times n^{C_p} \exp\{C_p n \max_{a \neq b} |R_{ab}|^4\}$, spherical density $\propto (\det R)^{(n-p-1)/2}$. At $p = 2$, $\det R = 1 - R_{12}^2$ and the exponent is $(n - 3)/2$, which is DH’s ρ .

What is new in the preprint.

- The cell comparison is for p replicas, with Gram boxes and a Legendre match to $-\frac{1}{2} \log \det R$ through cubic order, and holds up to overlaps of a small constant. DH treat $p = 2$ by Stirling's formula for $|q| = o(N^{-1/4})$.
- A field rotating with the matrix, a saddle off the edge ($d \asymp r$), and errors on the scale $k = nr^3$ with the matrix tilts of (6.2). DH work at zero field with a GOE average, BS at small β with no field.
- After the reduction, each direction, of strength $u = 1 \pm R_{12}$, is controlled by a one-direction bound $u^2/4$ proved with a Gaussian of precision $z - uJ$; the second direction is handled in the codimension-one compression (p. 22), where the bound holds unchanged by interlacing (§15.4, p. 118). BS instead replace V_2 by an independent V'_2 on a localization event (Lemma 3.2, (3.16)–(3.17), p. 11) and evaluate the rate by a large-deviation principle for V_1 (Prop. 3.4, Lemma 3.3, p. 12); DH use the log-concavity of K .
- The lower tail of F comes from controlled rotations (§6.2, pp. 23–25); BS use Lipschitz concentration over Haar measure (Lemma 3.1 with (3.12), pp. 9–10).

How to check.

1. Read the global pair estimate (p. 22, from “Only the pair calculation” to “proves the global pair estimate”) against BS §3.2.1 up to (3.14) (p. 10).
2. Read (6.3) and the five sentences after it (p. 21) against DH Lemma 3.9 (p. 12) and (2.10)–(2.11) (p. 7), setting $p = 2$.
3. Read $\mathbb{E}_U e^F = 1$, $\mathbb{E}_U e^{2F} \leq e^{o(k)}$ (pp. 22–23) against DH (2.3) (p. 6), Lemma 2.1 with (2.9) (p. 7) and the first display of the proof of Theorem 1.6 (p. 13).
4. For credit, read §1 (pp. 5–6), §7 (p. 26), §12.4 (p. 77) and §15.4 (p. 116) for DH, AcceptedCritical §5 (p. 22) for BS, and §3 (p. 7), where the preprint cites that companion only for its Theorems 1.2 and 1.3.

4.4 P4. Earlier critical-temperature bounds on the SK partition function and covariance (Du–Huang; El Alaoui–Gaitonde)

Summary. Three statements that the preprint proves at $\beta = 1$ have earlier versions in Du and Huang’s free-energy CLT and overlap bounds (DH below) [6, Thms 1.3–1.4, p. 3] and El Alaoui and Gaitonde’s critical covariance bound (EAG below) [18, Prop. 4.5, p. 11]. In the preprint they are the partition-function lower bound (7.5), the covariance lower bound (7.6) (both in §7.1, p. 29), and the bottom-scale covariance bound (12.8) of Proposition 12.4 (p. 77). The earlier version is stronger for (7.5) and weaker for (7.6). For (12.8) only an analogue at $s = 0$, under the ordinary law and with $\log n$ in place of ℓ^C , follows from DH’s Theorem 1.4(a). The preprint cites DH only for comparison methods (Theorem 1.6, §3.3, and Lemma 3.5 of their overlap paper [29]), never for Theorems 1.3 or 1.4, and never cites EAG; it attributes the zero-field in-probability covariance bounds to the companion AC, whose introduction credits DH’s Theorem 1.4. **Grade: Partial**, because the earlier forms of (7.6) and (12.8) are weaker or restricted to $s = 0$, and the stronger earlier form of (7.5) is credited only indirectly, through the comparison theorem from which it follows.

Setup. AC denotes the companion [24]. At zero field μ_J is invariant under $x \mapsto -x$. With $\Sigma_0 = \text{Cov}_{\mu_J}(X)$, $\langle \cdot \rangle$ the Gibbs average over independent replicas x^1, x^2 , and $R_{12} = (x^1)^\top x^2 / n$,

$$n^2 \langle R_{12}^2 \rangle = \|\Sigma_0\|_{\text{HS}}^2, \quad \|\Sigma_0\| \leq \|\Sigma_0\|_{\text{HS}} \leq (n\|\Sigma_0\|)^{1/2},$$

the last because $\text{Tr } \Sigma_0 = n$.

- DH write $N = n$ and $W = J$ (GOE with diagonal), and $Z_N = 2^{-N} \sum_x e^{x^\top W x / 2}$, which is $Z(0)$. Their $F_N = \log Z(0)$, $Z_N^{\text{sph}} = Z_o(0)$ and $X_N = Z_N / Z_N^{\text{sph}}$.
- EAG’s measure at $\beta = 1$, $\mu \propto \exp(n^{-1/2} \sum_{i < j} g_{ij} \sigma_i \sigma_j)$, is μ_J with $J_{ij}^\circ = g_{ij} / \sqrt{n}$. Their $\text{cov}(\mu) = \langle \sigma \sigma^\top \rangle$ is Σ_0 .
- In §12, $\Sigma_s = \text{Cov}_{\mu_s}(X)$, $\ell = \log \log(n + 1000)$, and $\mathcal{G}$ is the spectral restriction imposed throughout §12 (p. 66). An “integrated planted mass” is a probability under the planted law at time s .

Role in the argument. In the proof of Proposition 14.1 (p. 107), (7.5) and $\mathbb{E}Z(0) = e^{n/4}$ transfer every planted exception of route 1, of conditional probability q_J , to ordinary disorder at polynomial cost: $\mathbb{E}_{\text{ord}}[\mathbf{1}_{\{\mathcal{G}, Z(0)/\mathbb{E}Z(0) \geq n^{-c}\}} q_J] \leq n^C \mathbb{E}_{\text{pl}}[\mathbf{1}_{\mathcal{G}} q_J]$. Proposition 12.4 covers the bottom range $ns^{3/2} \leq K_{\log} \log n$, where the backward induction of §12 stops. In the same proof (p. 109) the variance decay $-V'_b \leq c_b V_b$ uses $c_b \leq Cn^{2/3} \ell^C$ for $b \leq b_* = (K'' \log n / n)^{2/3}$, with integral $o(\log n)$, inside $\int_0^T c_b db \leq \frac{2}{3}(1 + \epsilon) \log n + o(\log n)$. (7.6) is supplementary. With $\mathcal{D}(f_v) \leq 1$ for unit linear tests (p. 29), it reproves the lower bound $R_J \geq n^{2/3 - o(1)}$, which the main theorem takes from AC through Theorem 1.2 (p. 4).

The earlier result. DH (arXiv v2, July 2026) treat $\beta = 1$, zero field.

- *Theorem 1.3(b)* (p. 3): $(\frac{\log N}{6})^{-1/2} (F_N - \frac{N}{4} + \frac{\log N}{12}) \rightarrow \mathcal{N}(0, 1)$. It follows “directly” (p. 5; proof on p. 31) from Theorem 1.6 (p. 5), $\mathbb{E}[(X_N - 1)^2] \lesssim N^{-1/3}$, and the spherical CLT (eq. (1.6), p. 3). *Lemma 6.1* (p. 33) gives the lower tail $\mathbb{P}(F_N \leq \frac{N}{4} - \frac{\log N}{12} - t) \leq 2 \exp(-t^2 / \log N)$ for all $t \geq 0$. Lemma 5.5 (p. 29) is a cruder version.
- *Theorem 1.4* (p. 3): for a universal $c > 0$, (a) $\mathbb{E}(\exp(cN^{1/3} |R_{12}|)) \leq 2$ and (b) $\mathbb{E}\langle R_{12}^2 \rangle \geq cN^{-2/3}$. Part (a) is proved in §6.2 (p. 34) from Propositions 3.1–3.2 (p. 9) and the lower tail of X_N (Theorem 1.6 and Lemmas 6.1 and 6.3, pp. 33–34). Part (b) is a by-product of the cavity-method variance proof (§5.4, p. 32).

EAG (arXiv v3, November 2024; first posted December 2022): for every β , *Lemma 4.1* (p. 10) gives $\mathbb{E}\|\text{cov}(\mu)\|_{\text{op}} \geq n\mathbb{E}\langle R_{12}^2 \rangle$. *Proposition 4.5* (p. 11; announced in Theorem 1.2, p. 2) gives, at $\beta = 1$, $\mathbb{E}\|\text{cov}(\mu)\|_{\text{op}} \geq c(n/\log n)^{3/16}$. It combines Chatterjee’s bound $\mathbb{E}\langle |R_{12}|^3 \rangle \geq 1/(Ln)$ with a Talagrand overlap tail bound.

Correspondence.

Statement	Preprint	Earlier	Relation
Partition function	(7.5): $Z(0) \geq n^{-C} e^{n/4}$ with probability tending to one	DH Thm. 1.3(b); Lemma 6.1	Earlier is stronger: (7.5) for every $C > \frac{1}{12}$ (Thm. 1.3(b)), with failure at most $2n^{-(C-1/12)^2}$ (Lemma 6.1)

Its derivation	Prop. 7.1 (p. 26) at $h = 0$, attributed to DH Thm. 1.6 (p. 26); then (5.7), $Z_o(0) \geq n^{-C} e^{n/4}$	Thm. 1.6, then the spherical CLT (1.6) (pp. 5, 31)	Same two steps
Covariance, upper	(12.8): $\ \Sigma_s\ \leq Cn^{2/3}\ell^C$ outside integrated planted mass n^{-B} (on $\mathcal{G}$), for $ns^{3/2} \leq K_{\log} \log n$	DH Thm. 1.4(a)	Case $s = 0$, under the ordinary law, with $\log n$ for ℓ^C
Covariance, lower	(7.6): $\ \Sigma_0\ \geq n^{2/3-o(1)}$ with probability tending to one	DH Thm. 1.4(b): $\mathbb{E}\ \Sigma_0\ _{\text{HS}}^2 \geq cn^{4/3}$	Annealed Hilbert–Schmidt form at the same scale; (7.6) implies it up to $n^{o(1)}$. With EAG Lemma 4.1 it gives $\mathbb{E}\ \Sigma_0\ \geq cn^{1/3}$
Credit	Theorem 1.2 (p. 4) and §7.1 (p. 29) cite AC for $\ \Sigma_J\ \in [n^{2/3}/M_n, M_n n^{2/3}]$	EAG Prop. 4.5: $\mathbb{E}\ \Sigma_0\ \geq c(n/\log n)^{3/16}$ AC, Thm. 1.2 (p. 2); AC §1.2 (p. 3) credits DH Thms. 1.4 and 1.6	Operator norm, in expectation, exponent $\frac{3}{16}$ Indirect, for the zero-field in-probability bounds only

The two key formulas, side by side:

- *Covariance upper bound.* DH Theorem 1.4(a): put $Y = \langle e^{cn^{1/3}|R_{12}|} \rangle$. Since $y \mapsto \log^2(e + y)$ is concave on $[0, \infty)$, $c^2 n^{2/3} \langle R_{12}^2 \rangle \leq \log^2(e + Y)$, so

$$\|\Sigma_0\| \leq \|\Sigma_0\|_{\text{HS}} \leq c^{-1} n^{2/3} \log(e + Y), \quad \mathbb{P}(Y > n^B) \leq 2n^{-B}.$$

Preprint (12.8): $\|\Sigma_s\| \leq Cn^{2/3}\ell^C$ outside integrated planted mass n^{-B} .

- *Partition function.* DH Lemma 6.1 with $t = (C - \frac{1}{12}) \log n$: $\mathbb{P}(Z(0) < n^{-C} e^{n/4}) \leq 2n^{-(C-1/12)^2}$. Preprint (p. 29): $Z(0) = Z_o(0) \cdot Z(0)/Z_o(0)$, with the first factor at least $n^{-C} e^{n/4}$ by (5.7) and the second $1 + o(1)$ by Proposition 7.1.

What is new in the preprint.

- Proposition 12.4 holds at every small positive time, along the observation field, under the planted law, with ℓ^C in place of $\log n$. That precision is used: with $\log n$, the integral of c_b over $(0, b_*]$ on p. 109 would be of order $(\log n)^{5/3}$, not $o(\log n)$. The proof is a finite-replica p -th moment expansion with label cancellation, not DH’s reweighted exponential-moment argument.
- (7.6) bounds the operator norm with probability tending to one at the sharp exponent, by conditioning one spherical edge mode and using the cube–sphere error of Proposition 7.1. DH’s bound is annealed and Hilbert–Schmidt, and EAG’s exponent is $\frac{3}{16}$.
- (7.5) adds nothing: it is weaker than DH’s Theorem 1.3(b) and is derived by the same two steps.

How to check.

1. Read (7.5) and the sentence before it (p. 29) and the opening of §7 (p. 26) against DH Theorem 1.3(b) (p. 3), the sentence after Theorem 1.6 (p. 5), and Lemma 6.1 (p. 33).

2. Read §12.4 and Proposition 12.4 (p. 77) against DH Theorem 1.4(a) (p. 3), and its use on p. 109.
3. Read (7.6) (p. 29) against DH Theorem 1.4(b) (p. 3), EAG Lemma 4.1 (p. 10) and EAG Proposition 4.5 (p. 11).
4. Read Theorem 1.2 (p. 4) and the opening of §7.1 (p. 29) against AC Theorem 1.2 (p. 2) and AC §1.2 (p. 3), where DH Theorems 1.4 and 1.6 are credited.

4.5 P5. Local convexity of the TAP free energy at positive observation times (Celentano)

Summary. Celentano [19, Thm. 2(i)(2) and (ii), pp. 10–11] proved that, in the planted model of El Alaoui–Montanari–Sellke (EAMS) at every inverse temperature $\lambda > 0$ and every observation strength $\gamma_0 > 0$, the EAMS TAP free energy is strongly convex on a ball of radius $\epsilon\sqrt{n}$ around a late AMP iterate and has a unique stationary point there, with probability tending to one. At $\lambda = 1$, $\gamma_0 = s$ this is a local form, for the critical planted posterior at time s , of the stability implication (13.1), which Proposition 13.4 (pp. 88–89) proves on the whole approximate-root region with probability $1 - e^{-an}$; the assertion recurs in Lemma 17.3 (p. 143) and Proposition 17.5 (p. 148). The preprint never cites Celentano. It cites Celentano–Fan–Mei (§1, p. 6; §8, p. 30) only for the conditional Kac–Rice organization at fixed signal, without side information (§8 calls their Lemma 4.8 and Appendix A.5 a “local-convexity refinement” of Fan–Mei–Montanari); the companion [25] credits Celentano only in its introduction, which the preprint does not cite for this. **Grade: Partial (B)**, because Celentano’s statement is a weaker, local version: one AMP-selected ball, a fixed- q Hessian, constants for one fixed γ_0 , and no rate.

Setup. Section 2 fixes the planted law and h_s . In §13 (p. 83), with $J^\circ = J - \text{diag } J$, $m = \tanh y$, $q(y) = \|m\|^2/n$, $b(y) = 1 - q(y)$, $V_y = \text{diag}(1 - m_i^2)$ and diagonal $A \geq 0$,

$$F_h(y) = y - h + (b(y)I - J^\circ)m, \quad \mathcal{H}(y, A) = I + A^{1/2} \left(b(y)I - J^\circ - \frac{2}{n} mm^\top \right) A^{1/2}.$$

Roots of F_h are TAP solutions, and $V_y^{-1/2} \mathcal{H}(y, V_y) V_y^{-1/2} = V_y^{-1} + b(y)I - J^\circ - \frac{2}{n} mm^\top$ is the Hessian of the TAP free energy with self-consistent q . The stability implication (13.1) is

$$\|F_h(y)\| \leq \rho_0 \sqrt{n}, \quad 0 \leq A \leq (1 + \eta_0)I, \quad n^{-1/2} \|A - V_y\|_F \leq \epsilon_A \implies \mathcal{H}(y, A) \succeq c_0 I.$$

Proposition 13.4 (pp. 88–89). Fix $0 < s_1 < T < \infty$. There are $\rho_0, \eta_0, \epsilon_A, c_0, c_*, a > 0$, depending on $[s_1, T]$, such that at every $s \in [s_1, T]$, under the planted law restricted to the spectral event, the probability that (13.1) fails at $h = h_s$, or that some y has $\|F_{h_s}(y)\| \leq \rho_0 \sqrt{n}$ and $q(y) < c_*$, is at most e^{-an} . This holds uniformly in s on compact intervals after decreasing the residual threshold. At small $r = \sqrt{s}$, failure of (13.1) has probability at most $e^{-A_* nr^3}$, for each fixed A_* .

The proof gauges X to $\mathbf{1}$, so $J = W + \mathbf{1}\mathbf{1}^\top/n$ and $h_s = s\mathbf{1} + \sqrt{s}g$, and works with J in place of J° (written $\tilde{F}_h, \tilde{\mathcal{H}}$) (p. 89) until it removes the diagonal at a cost $(1 + \eta_0) \max_i |J_{ii}|$ (p. 94). Route 2 states the same test as (17.2) (p. 140), with residual $|F_h(y)|_n \leq 2\rho$ and interaction $J_* = \beta W_{\text{off}}$; at $\beta = 1$ it is (13.1).

Role in the argument. The implication (13.1) is the stable-field hypothesis recalled in Section 3.1, and Proposition 13.4 is the preprint’s replacement at $\beta = 1$ for the companion’s stable-field statements, which assume $\beta < 1$ (p. 83). With Proposition 13.5 (p. 95), Proposition 13.4

gives $\|\Sigma_s\| \leq C$ at each fixed positive time outside probability e^{-cn} (p. 103). This is the input from which §12 amplifies covariance bounds backward along the observation path, and (13.1) with (13.2) controls changes of mean in the entropy localization, where the small-time rates are used (§14.2, p. 110). On Route 2, Lemma 17.3 proves (17.2) at $\beta = 1$ along $Y(t) = tX + B(t)$, $t \in [t_b, T]$, under typical disorder, for use in §19.4 (p. 178); its exponential rate is what carries it from the planted law to typical disorder, using only $\log Z - \log \mathbb{E}Z = o(n)$ (p. 145). Proposition 17.5 proves (17.2) with fixed slack at small fixed t , for the field family (17.7) (p. 145), which contains $tX + B(t)$; the stability step is on pp. 156–157. It is used to start the cap induction (§18.1, p. 159) and at small times in §19.6 (p. 181).

The earlier result. Celentano’s vectors and matrices are set in bold. The model [19, (1)–(2), p. 2] is

$$\mathbf{Y} = \frac{\lambda}{n} \mathbf{x} \mathbf{x}^\top + \mathbf{W}, \quad \mathbf{y} = \gamma_0 \mathbf{x} + \sqrt{\gamma_0} \mathbf{g},$$

with $\mathbf{W} \sim \text{GOE}(n)$, $\mathbf{x}$ uniform on $\{-1, 1\}^n$ and $\mathbf{g}$ standard Gaussian; λ is the inverse temperature of the SK measure $\mu_{\mathbf{W}} \propto e^{(\lambda/2)\langle \mathbf{x}, \mathbf{W} \mathbf{x} \rangle}$ (p. 3). With $Q(\mathbf{m}) = \|\mathbf{m}\|^2/n$ and h the binary entropy, the EAMS TAP free energy (eq. (6), p. 4) is

$$\mathcal{F}_{\text{AMS}}(\mathbf{m}) = -\frac{\lambda}{2n} \mathbf{m}^\top \mathbf{Y} \mathbf{m} - \frac{1}{n} \sum_i h(m_i) - \frac{1}{n} \langle \mathbf{y}, \mathbf{m} \rangle - \frac{\lambda^2(1 - q_\infty)(1 + q_\infty - 2Q(\mathbf{m}))}{4},$$

where $q_\infty = \mathbb{E} \tanh^2(\gamma_\infty + \sqrt{\gamma_\infty} G)$ and $\gamma_\infty = \lambda^2 \mathbb{E} \tanh^2(\gamma_\infty + \sqrt{\gamma_\infty} G) + \gamma_0$ (eq. (15) and Prop. 3(v), p. 9). Let $\mathbf{m}^{k-1}$ be the $(k-1)$ st iterate of the EAMS AMP iteration started at $\mathbf{y}$ (eq. (5), p. 4). *Theorem 2* (pp. 10–11), case (i)(2): if $\lambda > 0$ and $\gamma_0 > 0$, there are $c, c' > 0$ depending on (λ, γ_0) such that, for every $\epsilon < c'$ and every k large enough (depending on $\lambda, \gamma_0, \epsilon$), with probability tending to one as $n \rightarrow \infty$,

$$\inf \{ \mathbf{v}^\top \nabla^2 \mathcal{F}_{\text{AMS}}(\mathbf{u}) \mathbf{v} : \|\mathbf{v}\| = 1, \|\mathbf{u} - \mathbf{m}^{k-1}\| \leq \epsilon \sqrt{n} \} > c/n \quad (\text{eq. (16)});$$

and (ii) for ϵ below a possibly smaller c' and k large, $\mathcal{F}_{\text{AMS}}$ has a unique stationary point in the ball $\|\mathbf{u} - \mathbf{m}^{k-1}\| \leq \epsilon \sqrt{n}$ intersected with $[-1, 1]^n$, lying in $(-1, 1)^n$. Celentano’s Hessian (p. 11), displayed with $\mathbf{W}$ and $\langle \mathbf{1}, \mathbf{v} \rangle$ after setting $\mathbf{x} = \mathbf{1}$ (§4), reads

$$n \nabla^2 \mathcal{F}_{\text{AMS}}(\mathbf{u}) = \text{diag}((1 - u_i^2)^{-1}) + \lambda^2(1 - q_\infty)I - \lambda \mathbf{Y}.$$

The motivating range is $\lambda \in [1/2, 1)$, but the theorem has no upper bound on λ : “We establish this for any $\lambda > 0$ ” (p. 4).

Correspondence. Take $\lambda = 1$ and $\gamma_0 = s$, and write $\Delta_{\mathbf{u}} = \text{diag}(1 - u_i^2)$ for $\mathbf{u} \in (-1, 1)^n$.

	Preprint ($\beta = 1$, time s)	Celentano ($\lambda = 1$, $\gamma_0 = s$)
Law	X uniform, $J = W + XX^\top/n$, $h_s = sX + B_s$; gauged to $X = \mathbf{1}$, $h_s = s\mathbf{1} + \sqrt{s}g$ (p. 89)	$\mathbf{Y} = \mathbf{x} \mathbf{x}^\top/n + \mathbf{W}$, $\mathbf{y} = s\mathbf{x} + \sqrt{s}\mathbf{g}$ ((1)–(2), p. 2); $\mathbf{x} = \mathbf{1}$ (p. 11)
Field equation	$\tilde{F}_{h_s}(y) = 0$: $y + (1 - q(y))m - Jm = h_s$ (p. 89)	AMP fixed point: $\mathbf{z} + (1 - Q(\mathbf{m}))\mathbf{m} - \mathbf{Y}\mathbf{m} = \mathbf{y}$, $\mathbf{m} = \tanh \mathbf{z}$ ((5), p. 4); stationarity of (6) has q_∞ for $Q(\mathbf{m})$

Overlap fixed point	$v_*(s) = s + \mathbb{E}_{N(v_*, v_*)} \tanh y$, with $q = v_* - s$ on the equality law (Lemma 13.3, p. 87)	$\gamma_\infty = s + \mathbb{E} \tanh(\gamma_\infty + \sqrt{\gamma_\infty} G)$, $q_\infty = \gamma_\infty - s$ ((15), Prop. 3(v), p. 9)
Matrix	$\Delta_{\mathbf{u}}^{-1/2} \tilde{\mathcal{H}}(y, \Delta_{\mathbf{u}}) \Delta_{\mathbf{u}}^{-1/2} = \Delta_{\mathbf{u}}^{-1} + b(y)I - J - \frac{2}{n} mm^\top$	$n \nabla^2 \mathcal{F}_{\text{AMS}}(\mathbf{u}) = \Delta_{\mathbf{u}}^{-1} + (1 - q_\infty)I - \mathbf{Y}$ (p. 11)
Where	every y with $\ F_{h_s}(y)\ \leq \rho_0 \sqrt{n}$ and every diagonal A in a Frobenius ball of radius $\epsilon_A \sqrt{n}$ around V_y ((13.1), p. 83)	every $\mathbf{u}$ with $\ \mathbf{u} - \mathbf{m}^{k-1}\ \leq \epsilon \sqrt{n}$, $\epsilon < c'$, $k \geq k_0(\epsilon)$ ((16), p. 10)
Constants, probability	margin c_0 ; constants depend only on $[s_1, T]$; probability $1 - e^{-an}$, uniformly on $[s_1, T]$; rate $e^{-A_* n r^3}$ for failure of (13.1), and $e^{-cn\theta^3}$ for approximate roots with $q(y) > \theta \geq Cr$, as $s \rightarrow 0$ (Prop. 13.4)	margin c (in units of $1/n$); c, c' depend on (λ, γ_0) ; probability tending to one
Roots	no approximate root with $q < c_*$; a unique root of F_{h_s} in $\mathbb{R}^n$ (proof of Prop. 13.5, p. 95)	a unique stationary point in the ball (Thm. 2(ii), p. 11)

The two key formulas, side by side:

- *Hessians.* With $J = \mathbf{Y}$ and $m = \tanh y$, for every y and $\mathbf{u} \in (-1, 1)^n$,

$$n \nabla^2 \mathcal{F}_{\text{AMS}}(\mathbf{u}) = \Delta_{\mathbf{u}}^{-1/2} \tilde{\mathcal{H}}(y, \Delta_{\mathbf{u}}) \Delta_{\mathbf{u}}^{-1/2} + (q(y) - q_\infty)I + \frac{2}{n} mm^\top.$$

Since $\Delta_{\mathbf{u}} \preceq I$, $\tilde{\mathcal{H}}(y, \Delta_{\mathbf{u}}) \succeq c_0 I$ gives $n \nabla^2 \mathcal{F}_{\text{AMS}}(\mathbf{u}) \succeq (c_0 - |q(y) - q_\infty|)I$.

- *Roots.* If $\mathbf{m}_* = \tanh \mathbf{z}_*$ is stationary for $\mathcal{F}_{\text{AMS}}$, then $\tilde{F}_{\mathbf{y}}(\mathbf{z}_*) = (q_\infty - Q(\mathbf{m}_*))\mathbf{m}_*$, and $\|\Delta_{\mathbf{u}} - V_{\mathbf{z}_*}\|_F \leq 2\|\mathbf{u} - \mathbf{m}_*\|$.

Hence, on the event of Proposition 13.4 and for n large, the lower bound in (16), centred at $\mathbf{m}_*$ instead of $\mathbf{m}^{k-1}$, holds with $c = c_0 - |Q(\mathbf{m}_*) - q_\infty| - o(1)$ on the ball $\|\mathbf{u} - \mathbf{m}_*\| \leq \frac{1}{2}\epsilon_A \sqrt{n}$ around every stationary point with $|Q(\mathbf{m}_*) - q_\infty| \leq \rho_0/2$ (the diagonal of J costs $o(1)$ in both the residual and the Hessian); for Celentano's stationary point this difference is small, since $Q(\mathbf{m}^{k-1}) \rightarrow q_\infty$ (Prop. 3, p. 9). Celentano proves the bound only around the stationary point that AMP finds.

What is new in the preprint.

- Uniformity over every approximate root (which gives a unique root of F_{h_s} in $\mathbb{R}^n$) and every nearby diagonal A , and exclusion of approximate roots of overlap below c_* ; Proposition 13.5 uses both.
- The Hessian with self-consistent q , including the rank-one term $-\frac{2}{n} mm^\top$.
- Probability $1 - e^{-an}$, uniform in $s \in [s_1, T]$ and along the path; small-time rates; transfer to typical disorder (Lemma 17.3); the spectral-noise fields of Proposition 17.5.
- The proof: smoothed Kac–Rice integrals, a conditional finite-rank Hessian and a local-volume witness, adapted from AcceptedGap §5. Celentano uses a Sudakov–Fernique comparison conditional on the AMP iterates.

How to check.

1. Read Celentano (1)–(2) (p. 2), (6) (p. 4) and Theorem 2 (pp. 10–11) with $\lambda = 1$, $\gamma_0 = s$, together with the sentence on p. 4 that covers every $\lambda > 0$, against Proposition 13.4 (pp. 88–89)

and the gauged planted law at the start of its proof (p. 89); then Celentano’s (15) and Prop. 3(v) (p. 9) against Lemma 13.3 (p. 87), which give $\gamma_\infty = v_*(s)$ and $q_\infty = v_*(s) - s$.

2. Put Celentano’s Hessian (p. 11) next to $\tilde{\mathcal{H}}$ (p. 89) conjugated by $\Delta_{\mathbf{u}}^{-1/2}$, and check the Hessian identity above; then compare the quantifiers of (16) (p. 10) with those of (13.1) (p. 83).
3. For Route 2, read Lemma 17.3 (p. 143) and the stability step of Proposition 17.5 (pp. 156–157) against Theorem 2 at $\gamma_0 = t$.
4. For credit, read the opening of §13 (p. 83), which introduces Proposition 13.4 as the end-point replacement for the companion’s stable-field statements; §3 (p. 7), which sets the whole-region assertion against “positivity at one selected root”; and the first sentence of the proof of Lemma 17.3 (p. 143). None cites Celentano; the companion credits him at AcceptedGap §1.1 (p. 4).

4.6 P6. Large-field reduction to sparse principal submatrices (Bandeira–El Alaoui–Rödger)

Summary. The preprint reuses a mechanism of Bandeira, El Alaoui and Rödger (BER below [20, Thm. 1.1 and proof of Lemma 2.3]: a large external field leaves only a small, configuration-dependent set of weakly biased coordinates, and a norm bound holding uniformly over all sparse principal submatrices makes the Gibbs measure effectively high-temperature on that set. It is used twice, each time to carry a near-critical posterior to a product law at a large observation time: in the stability verification of §19.4 (p. 178), and in the proof of Proposition 13.6 (pp. 102–103). BER is cited nowhere: §19.4 cites nothing, and §13.4 cites the companion’s net lemma, whose lead-in paragraph (companion §8, p. 61) credits BER. **Grade: Partial**, because only BER’s two elementary steps (the weak-field count and the sparse-submatrix bound) are reused, and the decisive block split is the companion’s.

Setup. AG denotes the companion [25]. In §§17–19, $\beta < 1$ is not an inverse temperature but a fixed interaction coefficient close to 1 (§18, pp. 158–159), and $|u|_n = n^{-1/2}\|u\|$.

- *Stability test.* For an interaction J_* with entries of variance j/n , (17.1) sets $m = \tanh y$, $r = 1 - \|m\|^2/n$, $V = \text{diag}(1 - m_i^2)$ and $F_h(y) = y + (jrI - J_*)m - h$, whose roots are TAP solutions. A field h passes the test (17.2) if every y and diagonal A with $|F_h(y)|_n \leq 2\rho$, $0 \leq A \leq (1 + \eta_{\text{stab}})I$ and $n^{-1/2}\|A - V\|_F \leq \epsilon_A$ satisfy

$$I + A^{1/2}(jrI - J_* - \frac{2j}{n}mm^\top)A^{1/2} \succeq bI.$$

At $A = V$ this is the symmetrized TAP Jacobian. For such fields with $j(1 + \eta_{\text{stab}})^2 < 1$, Proposition 17.1 uses AG’s high-temperature machinery to give a unique root and $\|\text{Cov}_{\mu_h}(x)\| \leq C$.

- *Weakening path (§19.4).* Route 2 tracks the retained entropy $H(s) = \mathbb{E}_{Q_\nu} D(\nu_s \| \mu_s)$, the expected relative entropy between the posteriors of the tilted law $\nu = f^2 \mu_J / N$ and of μ_J (§19.2, p. 173). After scalar precision T it follows $B_a = (T + 4a)I + (1 - \beta + a)J$, $0 \leq a \leq \beta$ (p. 178). The posterior interaction is $J_* = (\beta - a)J^\circ$, so $j = (\beta - a)^2$ falls to 0, a product law. The field is $h = TX + [4aX + (1 - \beta + a)JX] + B_a^{1/2}G$ with $X \sim \nu$, so every bound must be uniform in X .
- *Terminal extension (§13.4).* Proposition 13.6 sets $h_T = TX + \sqrt{T}g$, for any law of X , and continues with $h_{T+\lambda} = h_T + \lambda RX + R^{1/2}B_\lambda$, $R = J + c_+I \succeq 0$. It shows that μ_{h_T} has unscaled gap at least $1/2$ and that the laws $\mu_{h_{T+\lambda}, \lambda}$, where $\mu_{h, \lambda} \propto \exp(\frac{1-\lambda}{2}x^\top J^\circ x + h^\top x)$, keep bounded

covariance and inverse gap up to the product law at $\lambda = 1$, all outside probability e^{-cn} . Its proof bounds the error term $\mu_h[a^\top Ua + C|a|^\top Q|a|]$ of AG’s signed curvature inequality (Lemma 8.1), restated on p. 102; here $U = \tanh B$ entrywise, $Q_{ij} = U_{ij}^2$, $a_i = v_i d_i f$, and v_i is the conditional spin variance.

Role in the argument. Both routes end at a product law, where a two-point log-Sobolev inequality and tensorization bound the entropy. Reaching it with only fixed losses needs bounded covariance (route 2, through (17.2)) or a uniform gap (route 1), uniformly in the source law. Near the start of the §19 path $\sqrt{j} \approx 1$, so the crude bound $(1 + \eta_{\text{stab}})\|J_*\| \approx 2(1 + \eta_{\text{stab}})\sqrt{j}$ cannot give (17.2); the sparse bound can. In route 2 the verification yields (19.17), $\dot{H} \geq -CH - e^{-an}$, which carries the retained entropy (19.16) to the product endpoint when $L \leq \lambda n$ (p. 180); the same test on the later weakening interval of §19.6 defines the stopping time used for $\lambda n \leq L \leq Cn$, so both entropy ranges of Theorem 19.3 pass through it. In route 1, Proposition 13.6 supplies bounded covariance, the terminal gap and the order- n entropy extension in the proof of Proposition 14.1 (pp. 106, 109, 111).

The earlier technique. BER prove polynomial-time Glauber mixing for quadratic Gibbs measures, SK at every β included, once the external field is large [20, §1.1, Thm. 1.1 and Cor. 1.2, p. 3]. We write K for their interaction matrix; their m is a dimension. *Theorem 1.1:* if $\max_{|S| \leq \delta n} \|K_{SS}\| \leq \frac{1}{5}$ for some $\delta \in (0, 1)$, then for some θ independent of n , lazy Glauber dynamics for $\nu(x) \propto \exp(\frac{1}{2}x^\top Kx + \theta \mathbf{1}^\top x)$ mixes in polynomial time. The proof combines two observations.

- *Few weak coordinates.* A field $\theta \mathbf{1} + v$ with $\|v\| = O(\sqrt{n})$ (v comes from pinning, Proposition 3.2, p. 10) is below $\theta/2$ only on a small set, by a Chebyshev count (proof of Lemma 2.3, p. 7). A spin-flip injection then shows that most spins align with the field (Lemma 2.3, Corollary 2.4).
- *High temperature on small sets.* The remaining randomness lives on a small set: the free spins after pinning (§3.2, p. 11), or the set $\mathcal{D}$ where two replicas disagree, on which the halved replica difference is Gibbs with interaction $2K_{\mathcal{D}\mathcal{D}}$ (Lemma 2.1, p. 5). The hypothesis then bounds its covariance (eq. (11)).

The GOE matrix satisfies the hypothesis by Gaussian concentration and a union bound over supports, which BER call routine (§3.3, pp. 11–12). AG adapt the mechanism at fixed $j < 1$, crediting BER’s Theorem 1.1 just before its Lemma 8.2 (§8, p. 61). AG’s Lemma 8.2 proves the sparse bounds with 9^k nets, and the proof of its Proposition 8.3 counts weak-field sites, then splits the curvature form into a block on the weak set $S(x)$, controlled by the sparse bound, and cross and complementary blocks, controlled by the global norm times $2\sqrt{\eta}$ and η , respectively, where $\eta = \text{sech}^2(T/4)$.

Correspondence. The gauge $x \mapsto X \odot x$ maps the field TX to $T\mathbf{1}$ and preserves submatrix norms, so BER’s θ is the preprint’s T .

Step	Preprint	BER	AG
Large field, $O(\sqrt{n})$ perturbation	Proof of Prop. 13.6 (p. 103); §19.4	$\theta \mathbf{1} + v$ (Lemma 2.3); v from pinning (Prop. 3.2)	$TX + \sqrt{T}Z$ (Prop. 8.3)
Weak-field count	Norm count (p. 103); “small site fraction” (§19.4)	Chebyshev count (p. 7)	Independence count (p. 63)

Small, configuration- dependent set	Weak conditional field (p. 103); A not small (§19.4)	Free set (p. 11); disagreement set $\mathcal{D}$ (Lemma 2.2)	$S(x)$ (eq. (8.16))
Uniform sparse bound	9^k nets, $ \mathcal{S} \leq \alpha n$ (p. 102); in words in §19.4	Hypothesis of Thm. 1.1; GOE in §3.3	9^k nets (Lemma 8.2)
High temperature on the small set	Block split (p. 103); (17.2) via $\ J_*\ $ and $ A^{1/2}m _n$ (§19.4)	Exact restriction to $\mathcal{D}$ (Lemma 2.1, eq. (11))	Block split (p. 64)
Output	(19.17); gap $\geq \frac{1}{2}$ uniformly in λ (Prop. 13.6)	Polynomial mixing (Thm. 1.1)	Gap $\geq \frac{1}{2}$ at fixed $j < 1$ (Prop. 8.3)

The two key formulas, side by side:

- *Weak-field count.* BER (p. 7): $\|v\|^2 \geq |\mathcal{S}|(\theta/2)^2$ on $\mathcal{S} = \{i : h_i \leq \theta/2\}$, so $\|v\| \leq C_2\sqrt{m}$ and $\theta > 4C_2/\sqrt{\delta}$ give $|\mathcal{S}| \leq \delta m/4$. Preprint (p. 103): $\sup_\lambda \|h_{T+\lambda} - TX\| \leq C(1 + \sqrt{T})\sqrt{n}$, so for large T fewer than $\alpha n/2$ sites have $|h_{T+\lambda,i}| < T/2$.
- *High-temperature step.* BER (eq. (11), p. 11): $\|\text{Cov}(\nu_{2K_{\mathcal{S}\mathcal{S}}})\| \leq (1 - 2\|2K_{\mathcal{S}\mathcal{S}}\|)^{-1} \leq 5$ for $|\mathcal{S}| \leq \delta n$. AG (p. 64): $a^\top Ua \leq [\varepsilon + M(2\sqrt{\eta} + \eta)]q_f$; preprint (p. 103): $a^\top Ua + C|a|^\top Q|a| \leq (1 + C)\{\varepsilon + M(2\sqrt{\eta} + \eta)\}q_f$, where $q_f = \sum_i v_i(d_i f)^2$.

What is new in the preprint.

- At §19 the mechanism verifies (17.2), including its rank-one term, along a path on which the interaction coefficient varies, so that AG’s high-temperature input applies; at §13 it feeds AG’s curvature inequality. BER use it to bound correlation matrices under pinning.
- The small set is defined deterministically from the field or an approximate TAP root, for every configuration and source law; BER’s alignment holds in Gibbs probability.
- At §13 the conclusion is an $O(1)$ gap with failure probability e^{-cn} , extended in λ to a product law. For the planted posterior at time T , BER’s Corollary 4.1 (p. 13) gives only polynomial mixing.

How to check.

1. Read §19.4 (p. 178) against BER §1.2 (pp. 3–4) and the definition of $\mathcal{S}$ and the bound $|\mathcal{S}| \leq \delta m/4$ in the proof of BER’s Lemma 2.3 (p. 7): the “small site fraction” is BER’s count with $\theta = T$.
2. Read the proof of Proposition 13.6 (pp. 102–103) against BER §3.3 (pp. 11–12), AG’s Lemma 8.2 (pp. 61–63) and the proof of AG’s Proposition 8.3 (pp. 63–64), with AG’s credit to BER on p. 61.
3. Read (17.2) and the end of the stability verification in §19.4 against AG’s block split (p. 64), then BER’s Lemma 2.1 (p. 5) and eq. (11) (p. 11): BER restrict exactly to the small set, while AG and the preprint down-weight its complement.

5 Structural finding

5.1 C1. The spectral-cap route is not needed for the log-Sobolev exponent

Summary. Route 1 ends with the support profile of Theorem 14.2 (p. 111). By the Hermon–Peres characterization of the log-Sobolev constant through Dirichlet eigenvalues of sets [21, Thm 1.2, p. 3 of arXiv v3], this profile alone gives $C_{\text{LS},J} < 17c_\omega^{-1}n^{2/3+\omega}$; Diaconis–Saloff-Coste [22, Cor. 3.11, p. 728], applied to the L^2 bound in the proof of Lemma 20.1 (p. 184), gives the same exponent up to a factor $\log n$. The preprint instead takes the upper bound on $C_{\text{LS},J}$ in Theorem 1.1 only from Theorem 19.3, the endpoint of the spectral-cap route §§15–19 (pp. 112–183), and states in §2 (p. 7) that this inequality “will require its own entropy and layer argument”. It cites Diaconis–Saloff-Coste only for Theorems 3.5 and 3.7, and does not cite Hermon–Peres. Conversely, Goel–Montenegro–Tetali [37, Lemma 4.2 and the remark after it, p. 15 of arXiv v1], also cited, turn Theorem 19.3 back into the support profile, so the two routes end at statements equivalent up to constant factors. Grade: MAJOR, because a component of the main theorem becomes a citation and about 70 pages become unnecessary for Theorem 1.1.

Setup. Shared notation is in Section 2. Write $h_t^x = e^{t\mathcal{L}_J}(x, \cdot)/\mu_J$ for the density at per-site time t from the start x , and

$$\tau_2 = \inf\{t > 0 : \max_x \|h_t^x - 1\|_{L^2(\mu_J)} \leq 1/e\}$$

for the worst-start L^2 mixing time of $e^{t\mathcal{L}_J}$. Lemma 20.1 assumes (20.1): $\mathcal{D}(f) \geq \alpha \mu(f^2) \log(1/p)$ for every $f \geq 0$ with $0 < p = \mu(f > 0) \leq \frac{1}{2}$. On the event of Theorem 14.2 this holds with $\mu = \mu_J$ and $\alpha = c_\omega n^{-2/3-\omega}$. Its proof follows $V_t = \mu_J((h_t^x)^2) = 1 + \|h_t^x - 1\|_2^2$ and uses $M = \max\{8, \mu_{J,\min}^{-1}\}$, where $\mu_{J,\min} \geq e^{-(3+\log 2)n}$ on $\|J\| \leq 3$ (20.5). The sources’ constants are written α_{HP} , α_{DSC} and ρ_{GMT} , distinct from the preprint’s α and the slack ρ of Theorem 19.3.

Role in the argument. The mixing upper bounds come from Lemma 20.1 with Theorem 14.2 (p. 185). In §20.3 (p. 186) the median consequence of Theorem 14.2 gives the upper bound on R_J , and Theorem 19.3 alone is named for the upper bound on $C_{\text{LS},J}$. Route 2 exists to prove Theorem 19.3. It comprises the cube–sphere comparison on caps (§15), the radial-deficit induction (§16), its initialization (§§17–18), and entropy retention closed by Lemma 19.1 (§19). Elsewhere Theorem 19.3 serves only as a second proof of the mixing bounds in §20.2 (p. 185) and, through $C_{\text{EP},J} \leq C_{\text{LS},J}/4$, in Corollary 20.2. Route 1 does not use route 2: §§5–14 refer to no result of §§15–19, and §20.1 uses only Theorem 14.2 and (20.5).

The earlier results.

- *Hermon–Peres.* For a finite irreducible chain (Ω, P, π) with $Q = (P + P^*)/2$, set $\mathcal{E}(f) = \langle (I - Q)f, f \rangle_\pi$, $c_{\text{LS}} = \inf\{\mathcal{E}(f)/\text{Ent}_\pi(f^2) : f \text{ non-constant}\}$ and $t_{\text{LS}} = 1/c_{\text{LS}}$ ((3.11), p. 9). For $A \subsetneq \Omega$, $\lambda(A)$ is the smallest eigenvalue of $I - Q_A$, with Q_A the restriction of Q to A (p. 3); by the Rayleigh quotient, $\lambda(A) = \min\{\mathcal{E}(f)/\pi(f^2) : f \neq 0, \text{supp } f \subset A\}$. Con_δ is the set of connected A with $\pi(A) \leq \delta$ (p. 2). With

$$\alpha_{\text{HP}} = \min_{A \in \text{Con}_{1/2}} \frac{\lambda(A)}{|\log \pi(A)|}, \quad \kappa = 1/\alpha_{\text{HP}} \quad (\text{HP (1.5)}),$$

Theorem 1.2 (p. 3) states, for every irreducible chain on a finite state space,

$$\kappa \leq t_{\text{LS}} \leq 2(\kappa + t_{\text{rel}}(1 + \log 49)) \leq 2\left(1 + \frac{1 + \log 49}{\log 2}\right)\kappa < 17\kappa \quad (1.7),$$

where t_{rel} is the inverse of the smallest nonzero eigenvalue of $I - Q$, and the third step uses $t_{\text{rel}} \log 2 \leq \kappa$ (1.6).

- *Diaconis–Saloff–Coste*. For an irreducible Markov kernel K , $H_t = e^{-t(I-K)}$ (p. 695), $h_t^x = H_t^x/\pi$, $\mathcal{E}(f, f) = \frac{1}{2} \sum_{x,y} |f(x) - f(y)|^2 K(x, y) \pi(x)$ and $\alpha_{\text{DSC}} = \min \mathcal{E}(f, f) / \text{Ent}_\pi(f^2)$ ((1.4)–(1.7), p. 696; DSC write the entropy as $\sum |f|^2 \log(|f|^2 / \|f\|_2^2) \pi$). Corollary 3.11 (p. 728): if (K, π) is reversible and $\tau = \inf\{t > 0 : \sup_x \|h_t^x - 1\|_2 \leq 1/e\}$, then $\frac{1}{2\alpha_{\text{DSC}}} \leq \tau \leq \frac{1}{4\alpha_{\text{DSC}}} (4 + \log \log \frac{1}{\pi_*})$.
- *Goel–Montenegro–Tetali*. For the same $\mathcal{E}$, $\rho_{\text{GMT}} = \inf \mathcal{E}(f, f) / \text{Ent}_\pi(f^2)$ (p. 15) and $\lambda_0(A) = \inf_{f \in c_0^+(A)} \mathcal{E}(f, f) / \pi(f^2)$, over nonnegative f supported in A (defined p. 4; variational form p. 9). Lemma 4.2 gives $\Lambda(r) \geq \rho_{\text{GMT}} \log(1/r) / (1 - r)$ for the spectral profile Λ , and the remark after it gives $\lambda_0(A) \geq \rho_{\text{GMT}} \log(1/\pi(A))$ (p. 15).

How the earlier results apply. Take $K = P = P_J$ in all three sources. Each Q_i is an orthogonal projection in $L^2(\mu_J)$ (p. 184), so P_J is reversible; it is irreducible because every single-spin flip has positive heat-bath probability. The objects then match as follows.

Sources	Preprint, with $K = P = P_J$
$\mathcal{E}(f) = \langle (I - Q)f, f \rangle_\pi$ (HP p. 9, before (3.11)); $\mathcal{E}(f, f)$ (DSC p. 696, after (1.4); GMT p. 3)	$\mathcal{D}_J(f)/n$, since $\langle (I - Q_i)f, f \rangle_{\mu_J} = \mu_J[\text{Var}_{\mu_J}(f x_{-i})]$ and $P_J = n^{-1} \sum_i Q_i$ ((1.2), p. 3)
c_{LS} (HP), α_{DSC} , ρ_{GMT} ; t_{LS}	$1/(nC_{\text{LS},J})$, with $C_{\text{LS},J}$ as in (1.4); $t_{\text{LS}} = nC_{\text{LS},J}$
$\lambda(A)$ (HP), $\lambda_0(A)$ (GMT)	$\min\{\mathcal{D}_J(f)/(n\mu_J(f^2)) : f \neq 0, \text{ supp } f \subset A\}$
$\alpha_{\text{HP}} = \min_{\text{Con}_{1/2}} \lambda(A) / \log \pi(A) $ (HP (1.5))	$\geq c_\omega n^{-5/3-\omega}$ by (14.16)
$H_t = e^{-t(I-K)}$; τ (DSC pp. 695, 728)	$e^{(t/n)\mathcal{L}_J}$; $\tau = n\tau_2$
π_* (DSC)	$\mu_{J,\min} \geq e^{-(3+\log 2)n}$ (20.5)

1. *Hermon–Peres*. If $\mu_J(A) \leq \frac{1}{2}$ and $f \neq 0$ is supported in A , then $p \leq \mu_J(A)$, so (14.16) gives $\mathcal{D}_J(f)/n \geq c_\omega n^{-5/3-\omega} \log(1/\mu_J(A)) \mu_J(f^2)$. Hence $\lambda(A) \geq c_\omega n^{-5/3-\omega} |\log \mu_J(A)|$ for every such A , connected or not, and $\alpha_{\text{HP}} \geq c_\omega n^{-5/3-\omega}$. Then (1.7) gives $nC_{\text{LS},J} < 17\kappa \leq 17c_\omega^{-1} n^{5/3+\omega}$ on the event of Theorem 14.2, where (14.16) holds for all f at once. With $\omega = \delta/2$ this is $C_{\text{LS},J} \leq n^{2/3+\delta}$ with probability tending to one, for every $\delta > 0$.
2. *Diaconis–Saloff–Coste*. Corollary 3.11 reads $nC_{\text{LS},J}/2 \leq n\tau_2$, that is $C_{\text{LS},J} \leq 2\tau_2$. The proof of Lemma 20.1 shows, for every start, $V_t \leq 8$ once $t \geq t_0 = \alpha^{-1} \log \frac{\log(M/4)}{\log 2}$ and $V_{t_0+t} - 1 \leq 7e^{-2\alpha(\log 2)t}$ (p. 184). So $\tau_2 \leq t_0 + \frac{2+\log 7}{2\alpha \log 2}$, and (20.5) gives $C_{\text{LS},J} \leq 2\alpha^{-1}(\log(6n) + 3) = O(n^{2/3+\omega} \log n)$.
3. *Converse*. With $\rho_{\text{GMT}} = 1/(nC_{\text{LS},J})$, the remark after GMT Lemma 4.2, applied to $|f|$ with $A = \{f \neq 0\}$, gives $\mathcal{D}_J(f) \geq C_{\text{LS},J}^{-1} \log(1/p) \mu_J(f^2)$. So Theorem 19.3, $C_{\text{LS},J} \leq n^{2/3+\rho}$, yields (14.16) with $n^{-2/3-\rho}$ in place of $c_\omega n^{-2/3-\omega}$.

The two endpoints side by side:

Preprint	With the earlier results
Route 1, (14.16), p. 111: $\mathcal{D}_J(f) \geq c_\omega n^{-2/3-\omega} \log(1/p) \mu_J(f^2)$ for $p \leq \frac{1}{2}$	HP (1.5) with $\mathcal{E} = \mathcal{D}_J/n$: $\alpha_{\text{HP}} = \min_A \frac{\lambda(A)}{ \log \pi(A) } \geq c_\omega n^{-5/3-\omega}$
Route 2, (19.23), p. 183: $\text{Ent}_{\mu_J}(f^2) \leq n^{2/3+\rho} \mathcal{D}_J(f)$, after §§15–19	HP (1.7): $C_{\text{LS},J} = t_{\text{LS}}/n < 17c_\omega^{-1} n^{2/3+\omega}$ from route 1 alone

What the cap route still contributes.

- A second proof of all the upper bounds in Theorem 1.1, through hypercontractivity (§20.2), that does not pass through Theorem 14.2.
- Results on capped Gibbs measures that nothing outside §§15–19 uses: the cube–sphere comparison of capped partition functions and projected means, uniformly over field balls (Proposition 15.1); the one-step transport of the radial deficit (Proposition 16.1, p. 125); and the projected covariance bounds $(1+\epsilon)/p^2$ on the saddle annulus and C/p^2 on bounded field balls at all scales $p_{\min}(n) \leq p \leq p_{\text{top}}$ (Theorem 18.6, p. 170).
- No better constant: Theorem 19.3 passes through Lemma 19.1, whose constant carries $C(1 + \log_+ M)^2 = O(\log^2 n)$ (p. 172), while the Hermon–Peres derivation above loses only the factor 17.

How to check.

1. *Hermon–Peres.* Put (14.16) (p. 111) beside HP (1.5) and Theorem 1.2 (p. 3), with c_{LS} from (3.11) (p. 9). With $P = P_J$, (1.2) gives $\mathcal{E} = \mathcal{D}_J/n$, and the factor n cancels between t_{LS} and κ .
2. *Diaconis–Saloff–Coste.* Put the L^2 bounds in the proof of Lemma 20.1 (p. 184) and (20.5) (p. 185) beside DSC Corollary 3.11 (p. 728) and the definitions (1.4)–(1.7) (p. 696).
3. *Use of route 2.* Read §20.3 (p. 186) and the sentence quoted from §2 (p. 7); then check that the proofs of Theorem 14.2 (pp. 111–112) and Lemma 20.1 (pp. 184–185) invoke nothing from §§15–19.
4. *Converse.* Put GMT Lemma 4.2 and the remark after it (p. 15) beside Theorem 19.3 (p. 183).

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